NETI ACADEMY

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NETI PRELIMS

Environment & Ecology

Current Affairs Magazine

COVERING OCTOBER 2024 — MARCH 2026

For UPSC Civil Services Prelims 2026 · 24 May 2026

The magazine that tells you what to study — and what to skip.

Edition 1.1 · April 2026 · Revised

A NOTE FROM NETI

How to use this magazine

You are holding the first edition of Neti Prelims — Environment & Ecology. It covers the eighteen months of environment current affairs most likely to be tested in UPSC Prelims 2026: October 2024 to March 2026. April and May will be covered in a separate supplement released closer to the exam.

This magazine is built on one idea: you do not need more information. You need less information, presented better, with clear signals on what matters most. Every page has earned its place. Every topic connects to a static source you should already be revising. Every fact is verified.

The priority system

Every topic carries one of three priority labels. Read them in this order:

- **MUST-KNOW:** High probability of direct or indirect question. These are non-negotiable.
- **SHOULD-KNOW:** Moderate probability. Read carefully; do not skip.
- **GOOD-TO-KNOW:** Low probability but useful for elimination. Skim if short on time.

What each topic contains

- **A 3–5 line factual summary** — what happened, plainly stated.
- **Key facts for Prelims** — the exact details UPSC is likely to test.
- **REVISE** — the static chapter/book you should revise alongside this.
- **PYQ LINK** — how this connects to a past question, where relevant.

Suggested reading calendar

Give this magazine ten days. That is enough. Twelve days if you also want to attempt the end-of-section MCQs and the Grand Practice Test. The calendar below is a suggestion — adjust to your pace.

Days	Section	Focus
Day 1	International Conventions & COPs	COP29, COP30, CBD COP16, UNCCD COP16, Ramsar COP15
Day 2	Climate Change & India's Commitments	NDC, Mission LiFE, Green Credit, Carbon Market
Day 3–4	Biodiversity & Species in News	IUCN updates, Project Cheetah, invasive species
Day 5	Protected Areas & Conservation	New Ramsar sites, Tiger Reserves, Biosphere Reserves
Day 6	Forests, Land & Soil	ISFR 2023, MISHTI, Aravalli Green Wall
Day 7	Pollution & Waste	CAQM, GRAP, Plastic Rules, EPR
Day 8	Renewable Energy & Clean Tech	Green Hydrogen, PM Surya Ghar, ISA, Nuclear
Day 9	Reports, Indices & Governance	Emissions Gap, Living Planet, EPI, State of Climate
Day 10	Quick-fire revision + Top 30 Facts	Final consolidation

Days	Section	Focus
Day 11–12	Grand Practice Test (25 MCQs)	Self-assessment and answer review

> *Most current affairs sources give you 1,000 pages and three months of reading. You finish nothing. Neti Prelims gives you only what matters — prioritized, connected to your syllabus, and built for recall. Ten days. That is enough.*

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07	Renewable Energy & Clean Tech	*5 topics*
08	Reports, Indices & Governance	*6 topics*
—	Top 30 Must-Know Facts	*Quick revision*
—	Grand Practice Test	*25 MCQs + Answer Key*

Environment has appeared in every UPSC Prelims paper from 2011 to 2025 without exception — averaging 13–14 questions per year, never below 10. It is the most reliable scoring subject in the exam. Invest here.

SECTION 01

International Conventions & COPs

The global climate and biodiversity calendar, 2024–2026

UPSC tests international environmental conventions every single year — never fewer than two questions, often four. It tests three things: which convention covers what, whether India is a party, and what each latest COP actually decided. The eight topics in this section cover every major COP outcome from October 2024 to March 2026. If you know these well, you have locked down three to four marks on exam day.

MUST-KNOW

1.1 COP30 Belém, Brazil — the Global Mutirão Decision

Prelims & Mains · International Affairs + Environment · Also: SSC, State PCS, CDS

The 30th Conference of the Parties (COP30) to the UNFCCC met in Belém, Brazil, from 10–21 November 2025 (concluding 22 November after overnight talks). Nearly 200 countries adopted the Belém Package, whose flagship outcome is the Global Mutirão Decision — a Portuguese word of Tupi-Guarani origin meaning 'collective effort.' The COP marked the tenth anniversary of the Paris Agreement.

COP30 was positioned as the 'COP of implementation' — the shift from pledges to action. But its central failure was the absence of any roadmap to phase out fossil fuels, despite support from over 80 countries. The final text only 'recalls' the COP28 UAE Consensus on 'transitioning away from fossil fuels.' Leaders of India, China, and the United States did not attend; the US did not send a delegation for the first time in COP history.

KEY FACTS FOR PRELIMS

Venue & dates	Belém, Pará, Brazil · scheduled 10–21 November 2025; talks concluded 22 November after overnight negotiations
Presidency	André Corrêa do Lago (Brazil); Brazil holds Presidency until Nov 2026
Flagship outcome	Global Mutirão Decision — 29 decisions adopted by consensus
Finance target	Mobilize USD 1.3 trillion annually by 2035 (Baku-to-Belém Roadmap)
Adaptation finance	Called to triple by 2035 (from ~USD 40 billion baseline)
New mechanisms	Global Implementation Accelerator · Belém Mission to 1.5°C · Belém Mechanism for Just Global Transition
Adaptation indicators	Belém Adaptation Indicators — 59 finalized for Global Goal on Adaptation
Fossil fuels	No phase-out roadmap in final text; Presidency-led initiative to continue
Host of COP31	Antalya, Türkiye (Nov 2026); under the deal, Australia withdrew its bid in exchange for its minister presiding over the negotiations and hosting a pre-COP in the Pacific

REVISE: *Shankar IAS / PMF IAS Environment — Chapter on Climate Change and International Agreements. Read UNFCCC, Kyoto Protocol, Paris Agreement in sequence.*

PYQ LINK: *UPSC has asked about UNFCCC, Kyoto Protocol, and Paris Agreement provisions in 2014, 2016, 2022. COP decisions with their numerical targets (finance amounts, temperature goals) are standard testing territory.*

MUST-KNOW

1.2 COP29 Baku, Azerbaijan — the New Collective Quantified Goal

Prelims & Mains · International Affairs + Environment · Also: State PCS, CDS

COP29 met at Baku, Azerbaijan, from 11–24 November 2024 and was labelled the 'finance COP.' Its central decision was the New Collective Quantified Goal (NCQG) on climate finance — the successor to the earlier USD 100 billion per year target. Developed countries committed to take the lead in mobilizing at least USD 300 billion per year for developing countries by 2035, with a broader aspiration of USD 1.3 trillion per year from all sources by 2035. The USD 300 billion figure was widely criticized by developing countries — India called the outcome 'optical illusion' and 'abysmally poor.'

COP29 also finalized the rules for Article 6 of the Paris Agreement — the mechanism for carbon market cooperation between countries (Article 6.2 for bilateral transfers of ITMOs; Article 6.4 for a UN-supervised central market). The Baku Adaptation Roadmap was launched, and the Loss and Damage Fund (agreed at COP27) became operational.

KEY FACTS FOR PRELIMS

Venue & dates	Baku, Azerbaijan · 11–24 November 2024
Theme	The 'Finance COP'
NCQG decision	USD 300 billion/year by 2035 from developed countries; USD 1.3 trillion/year from all sources
Article 6 rules	Article 6.2 (bilateral ITMOs) and Article 6.4 (centralized UN market) operational
Loss & Damage Fund	Became operational; first board meeting held in Baku
India's stance	Rejected the quantum as inadequate; reiterated CBDR-RC and need for grant-based finance
ITMO definition	Internationally Transferred Mitigation Outcomes — tradable emission reductions between Parties

REVISE: *Shankar IAS / PMF IAS — Paris Agreement chapter. Supplement with a quick note on CBDR-RC (Common But Differentiated Responsibilities and Respective Capabilities).*

PYQ LINK: *UPSC 2023 Mains asked about India's stance on climate negotiations. Prelims has tested Paris Agreement provisions repeatedly. NCQG terminology and dollar figures are exactly the type of fact UPSC tests in Statement-Based questions.*

MUST-KNOW

1.3 CBD COP16 Cali + Rome — the Cali Fund and DSI Mechanism

Prelims & Mains · Environment + International Affairs · Also: State PCS

The 16th Conference of the Parties to the Convention on Biological Diversity (CBD) met in two parts. The first session was held in Cali, Colombia, from 21 October – 2 November 2024; it was suspended before all decisions could be adopted because quorum was lost. The resumed session took place in Rome in February 2025, where remaining decisions were finalized and the Cali Fund was officially launched.

The central outcome was the operationalization of a global mechanism for benefit-sharing from Digital Sequence Information (DSI) on genetic resources — known as the Cali Fund. Large companies in pharmaceuticals, biotechnology,

cosmetics, and similar sectors that commercially use DSI are expected to contribute 0.1% of revenue or 1% of profits. At least 50% of the Cali Fund's resources must go to Indigenous Peoples and local communities (IPLCs).

COP16 also formally established a new permanent Subsidiary Body under Article 8(j) — giving Indigenous Peoples and local communities a permanent decision-making seat. It is the first such body under any UN biodiversity treaty.

KEY FACTS FOR PRELIMS

Convention	Convention on Biological Diversity (CBD, 1992) — one of the three Rio Conventions
Venues & dates	Cali, Colombia (Oct–Nov 2024) + Rome resumed session (Feb 2025)
Cali Fund	Global fund for fair sharing of benefits from DSI on genetic resources
Contribution rate	0.1% of revenue or 1% of profits from eligible large companies (assets >USD 20m, sales >USD 50m, profit >USD 5m)
IPLC share	At least 50% of Cali Fund resources earmarked for Indigenous Peoples and local communities
Article 8(j)	New permanent Subsidiary Body established — historic win for IPLCs
Framework guiding targets	Kunming-Montreal Global Biodiversity Framework (KMGBF, 2022) — 23 targets for 2030
GBFF Fund	Global Biodiversity Framework Fund received additional USD 163 million pledges at COP16

REVISE: *Shankar IAS / PMF IAS Environment — Chapter on Biodiversity Conservation → Convention on Biological Diversity, Nagoya Protocol (access & benefit sharing), Cartagena Protocol (biosafety).*

PYQ LINK: *UPSC 2014 asked about Nagoya Protocol. UPSC 2022 asked about the three objectives of CBD. DSI and Cali Fund are the 2024–25 evolution of the same theme — benefit sharing.*

MUST-KNOW

1.4 UNCCD COP16 Riyadh — land, drought, and the Riyadh Action Agenda

Prelims & Mains · Environment + International Affairs · Also: State PCS, CDS

The 16th session of the UN Convention to Combat Desertification (UNCCD) met in Riyadh, Saudi Arabia, from 2–13 December 2024. It marked the 30th anniversary of the Convention (adopted June 1994) and was the first UNCCD COP held in the Middle East and North Africa. With over 20,000 participants, it was the largest UNCCD COP ever.

Parties adopted 39 decisions but failed to agree on a legally binding drought protocol — the African Group had pushed hard for this. Instead, the groundwork was laid for a future global drought regime, to be finalized at COP17 in Mongolia (2026). Over USD 12 billion was pledged through the Riyadh Global Drought Resilience Partnership, including USD 10 billion from the Arab Coordination Group.

KEY FACTS FOR PRELIMS

Convention	UNCCD — one of the three Rio Conventions (with UNFCCC and CBD)
Adopted	1994 (30th anniversary at COP16); India ratified 1996
COP16 venue	Riyadh, Saudi Arabia · 2–13 December 2024
Key outcome	Riyadh Action Agenda — restore 1.5 billion hectares globally by 2030

Drought pledges	Riyadh Global Drought Resilience Partnership — USD 12.15 billion for 80 vulnerable countries
Financing need	UNCCD estimates USD 2.6 trillion needed by 2030 for land restoration
Drought protocol	Could not agree on legally binding drought regime — pushed to COP17 (Mongolia, 2026)
New tool launched	International Drought Resilience Observatory (IDRO) — first AI-driven global drought platform
New initiative	Rio Trio Initiative — aligns UNCCD, UNFCCC, and CBD processes

REVISE: *Shankar IAS / PMF IAS — Chapter on Land Degradation and Desertification. Note the three Rio Conventions (Earth Summit 1992).*

PYQ LINK: *UPSC 2016 asked to match Rio Convention with its year of adoption. UNCCD, UNFCCC, CBD distinctions are classic UPSC testing territory.*

SHOULD-KNOW

1.5 Ramsar COP15 Victoria Falls — India's Mission LiFE resolution

Prelims · Environment · Also: State PCS

The 15th Conference of the Parties to the Ramsar Convention on Wetlands was held in Victoria Falls, Zimbabwe, from 23–31 July 2025 — the first Ramsar COP ever held in Africa. India played a lead role, proposing and securing adoption of a resolution titled 'Promoting Sustainable Lifestyles for the Wise Use of Wetlands,' aligned with India's flagship Mission LiFE (Lifestyle for Environment) initiative.

At the time of COP15, India had 91 Ramsar sites; by early 2026 the figure has grown to 98, making India home to the highest number of Ramsar sites in Asia and third-highest globally (after the UK and Mexico). Tamil Nadu leads with 20 sites. Two Indian sites — Keoladeo National Park (Rajasthan) and Loktak Lake (Manipur) — remain on the Montreux Record of sites facing ecological change.

KEY FACTS FOR PRELIMS

Convention	Ramsar Convention on Wetlands (1971, Ramsar, Iran) — entered force 1975
India a party since	1 February 1982
COP15 venue & dates	Victoria Falls, Zimbabwe · 23–31 July 2025 (first Ramsar COP in Africa)
India's resolution	'Promoting Sustainable Lifestyles for the Wise Use of Wetlands' — linked to Mission LiFE
Total Ramsar sites (India)	98 (as of January 2026) — highest in Asia, third globally
Most recent sites	Patna Bird Sanctuary (UP) — 97th; Chhari-Dhand (Gujarat) — 98th
First Ramsar sites in India	Chilika Lake (Odisha) and Keoladeo NP (Rajasthan) — both 1981
Montreux Record (India)	Keoladeo NP and Loktak Lake (Chilka was removed in 2002)
World Wetlands Day	2 February (UNGA Resolution 75/317, 2021)

REVISE: *Shankar IAS / PMF IAS — Chapter on Wetlands + Ramsar Convention. Memorize the definition of wetlands (areas of marsh, fen, peatland or water, static or flowing, fresh, brackish, or salt, including marine water to 6 m depth at low tide).*

PYQ LINK: *UPSC 2019 asked about wetlands conservation: which of the following are Ramsar sites. UPSC 2020 asked about Montreux Record. Both are recurring themes.*

MUST-KNOW

1.6 Global Plastics Treaty — INC-5.1 Busan, INC-5.2 Geneva, INC-5.3

Prelims & Mains · Environment + Pollution · Also: State PCS, SSC

In March 2022, the UN Environment Assembly (UNEA-5.2) adopted Resolution 5/14 mandating an Intergovernmental Negotiating Committee (INC) to develop a legally binding instrument on plastic pollution, addressing the full life cycle of plastics. The original deadline was end-2024.

INC-5 Part 1 (Busan, Republic of Korea, November–December 2024) failed to finalize the treaty; a 'Chair's Text' was agreed as the basis for future negotiations. INC-5.2 (Geneva, 5–15 August 2025) again ended without consensus — divisions persisted between ~100 countries pushing for an ambitious treaty with binding production cuts (the 'High Ambition Coalition' led by Rwanda-Norway and including the EU) and oil-producing states (Saudi Arabia, Russia, Iran, and others) resisting production controls. INC-5.3 was held on 7 February 2026 but was procedural only — it elected a new Chair after Ecuador's Ambassador Vayas Valdivieso resigned. Substantive negotiations remain unfinished.

KEY FACTS FOR PRELIMS

Mandate	UNEA Resolution 5/14 (March 2022) — legally binding instrument on plastic pollution
Secretariat	UN Environment Programme (UNEP)
INC sessions	INC-1 Uruguay, INC-2 Paris, INC-3 Nairobi, INC-4 Ottawa, INC-5.1 Busan, INC-5.2 Geneva, INC-5.3 Geneva
INC-5.1	Busan, 25 Nov – 1 Dec 2024 — Chair's Text issued; no treaty
INC-5.2	Geneva, 5–15 August 2025 — no consensus; revised drafts rejected
INC-5.3	Geneva, 7 February 2026 — procedural only; new Chair elected
Core divide	Production limits (ambitious countries) vs waste management only (oil-producing nations)
Plastic pollution scale	~460 million tonnes of plastic produced annually; ~20 million tonnes leak into the environment
India's position	Supports national action plans and voluntary measures; opposes binding upstream production caps

REVISE: *Shankar IAS / PMF IAS — Pollution chapter → Plastic Waste Rules, 2016 (as amended). Note the Basel Convention amendment (2019) brought plastic waste trade under control.*

PYQ LINK: *UPSC 2021 asked about Single-Use Plastic ban in India. UPSC Mains 2018 asked about plastic pollution. Global treaty talks are the logical Prelims angle.*

SHOULD-KNOW

1.7 CMS COP14 & the State of Migratory Species

Prelims · Environment · Also: State PCS

Note: CMS COP14 (February 2024) falls slightly before this magazine's main coverage window (October 2024 – March 2026) but is retained here because its outcomes remain the current reference framework for migratory-species conservation through 2026, and UPSC typically tests such conventions on a 12–18 month horizon.

The 14th Conference of the Parties to the Convention on the Conservation of Migratory Species of Wild Animals (CMS) was held in Samarkand, Uzbekistan, from 12–17 February 2024. It released the first-ever State of the World's Migratory Species Report, which found that 44% of listed migratory species are experiencing population decline, and one in five are threatened with extinction.

New additions to CMS Appendix I (strictly protected, endangered) included the Eurasian Lynx, Sand Tiger Shark, and Balkan Lynx. The CMS Samarkand Strategic Plan for Migratory Species (2024–2032) was adopted. India hosted the previous CMS COP (COP13, Gandhinagar, 2020), where the 'Gandhinagar Declaration' on ecological connectivity was adopted.

KEY FACTS FOR PRELIMS

Convention	CMS / Bonn Convention (1979) — UN Environment Programme
Appendices	Appendix I — endangered migratory species (strict protection); Appendix II — species needing international cooperation
COP14	Samarkand, Uzbekistan · 12–17 February 2024
Major report	First State of the World's Migratory Species — 44% in decline; 22% threatened
India & CMS	India party since 1983; hosted COP13 in Gandhinagar, 2020 (Gandhinagar Declaration)
Central Asian Flyway	India has launched the National Action Plan for Conservation of the Central Asian Flyway (2018–2023; updated 2024)

REVISE: *Shankar IAS / PMF IAS — Chapter on International Conventions. Link CMS with other wildlife treaties: CITES (trade), IWC (whales), ICRW.*

PYQ LINK: *UPSC 2020 asked to match conventions with their subject matter. CMS migratory species categories are testable.*

SHOULD-KNOW

1.8 BBNJ Agreement, Basel amendments, and the Convention Landscape

Prelims · Environment + International Affairs · Also: State PCS

The BBNJ Agreement (Biodiversity Beyond National Jurisdiction) — also called the High Seas Treaty or 'Paris Agreement for the oceans' — was adopted in June 2023 under UNCLOS. It regulates marine biological diversity in areas beyond any country's exclusive economic zone (~64% of the ocean). As of March 2026, it is close to the 60 ratifications needed for entry into force. India signed in September 2024 but has not yet ratified.

The Basel Convention (1989) — on transboundary movement of hazardous wastes — adopted the Basel Plastic Amendments (effective 2021), bringing most plastic waste under the Prior Informed Consent procedure. India implements this via the Hazardous and Other Wastes (Management) Rules.

KEY FACTS FOR PRELIMS

BBNJ Agreement	High Seas Treaty under UNCLOS — June 2023; covers ABNJ (areas beyond national jurisdiction)
BBNJ four pillars	Marine Genetic Resources + benefit sharing · Area-Based Management Tools (incl. MPAs) · Environmental Impact Assessments · Capacity building & tech transfer
India's status	Signed September 2024; ratification pending
Basel (1989)	Hazardous waste movement; India party since 1992
Rotterdam (1998)	Prior Informed Consent for hazardous chemicals & pesticides
Stockholm (2001)	Persistent Organic Pollutants (POPs) — 'dirty dozen' chemicals
Minamata (2013)	Mercury — covers mining, emissions, and products
Vienna (1985) + Montreal (1987)	Ozone layer protection — Kigali Amendment (2016) added HFCs

REVISE: *Shankar IAS / PMF IAS — International Environmental Conventions chapter. Keep a one-page chart listing each convention, year, scope, and India's status.*

PYQ LINK: *UPSC 2014 asked about the Montreal Protocol. UPSC 2015 asked about Minamata. UPSC 2016 asked about the Cartagena Protocol. These are certainties — cover every major convention.*

SECTION 1 — PRACTICE

Test yourself

Q1. With reference to the COP30 held in Belém, Brazil (November 2025), consider the following statements:

(a) The flagship outcome was the Global Mutirão Decision, adopted by consensus of nearly 200 Parties.
(b) The final text contained a binding roadmap for the phase-out of fossil fuels.
(c) The adaptation finance was called to triple by 2035 relative to the earlier USD 40 billion goal.
(d) Brazil will host COP31 in 2026.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 2 and 4 only
(c) 1, 2 and 3 only
(d) 1, 3 and 4 only

Q2. The Cali Fund, launched under the Convention on Biological Diversity in 2024–25, is related to:

(a) Restoration of mangrove ecosystems in the Caribbean
(b) Fair sharing of benefits arising from the use of Digital Sequence Information on genetic resources
(c) Compensation to countries hosting UNESCO biosphere reserves
(d) Financing of National Biodiversity Strategy and Action Plans under the Montreal Protocol

Q3. Consider the following pairs of Ramsar sites in India with their locations:

(a) Khichan — Rajasthan
(b) Udaipur Jheel — Bihar
(c) Khecheopalri — Sikkim

(d) Chhari-Dhand — Gujarat

How many of the above pairs are correctly matched?

(a) Only two
(b) Only three
(c) All four
(d) Only one

Q4. Statement I: The UNCCD COP16 held at Riyadh (December 2024) adopted a legally binding protocol on drought. Statement II: The UNCCD is one of the three Rio Conventions that emerged from the 1992 Earth Summit.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Q5. With reference to the BBNJ Agreement (Biodiversity Beyond National Jurisdiction), consider the following statements:

(a) It was adopted in 2023 as an agreement under the UN Convention on the Law of the Sea (UNCLOS).
(b) It covers marine biodiversity in areas beyond national jurisdiction, which constitute over 60% of the ocean.
(c) India has ratified the BBNJ Agreement.
(d) One of its four pillars is the establishment of Marine Protected Areas (MPAs).

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1, 2 and 4 only
(c) 2, 3 and 4 only
(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (d) 1, 3 and 4 only

Statement 2 is incorrect — COP30 did not include a fossil fuel phase-out roadmap despite backing from 80+ countries; the Presidency launched a separate (non-treaty) Presidency-Initiative on this. Brazil's COP Presidency runs until November 2026, so India's next reference point is COP31 hosted by Türkiye at Antalya.

Q2. Answer: (b)

The Cali Fund, launched at CBD COP16 (Cali 2024 / Rome resumed session 2025), is the global fund for fair and equitable sharing of benefits from the use of Digital Sequence Information (DSI) on genetic resources. Large commercial users contribute 0.1% of revenue or 1% of profits; at least 50% goes to Indigenous Peoples and local communities.

Q3. Answer: (c) All four

All four are correctly matched. Khichan & Menar (Rajasthan, June 2025); Udaipur Jheel & Gokul Jalashay (Bihar, 2025); Khecheopalri (Sikkim, 2025); Chhari-Dhand (Gujarat) is India's 98th Ramsar site (early 2026).

Q4. Answer: (d)

Statement I is incorrect — UNCCD COP16 failed to adopt a binding drought protocol; parties could not agree and pushed the issue to COP17 (Mongolia 2026). Statement II is correct — UNCCD (1994) is one of the three Rio Conventions along with UNFCCC (1992) and CBD (1992). Note: UNCCD was negotiated after the Earth Summit but as a direct outcome of it.

Q5. Answer: (b) 1, 2 and 4 only

Statement 3 is incorrect — India signed the BBNJ Agreement in September 2024 but has not yet ratified it. The Agreement covers ~64% of the ocean (ABNJ), was adopted in 2023 under UNCLOS, and its four pillars include MGRs with benefit sharing, Area-Based Management Tools (including MPAs), EIAs, and capacity building.

SECTION 02

Climate Change & India's Commitments

Targets, instruments, and the architecture of climate action

Climate change is a recurring theme — not just internationally, but in terms of India's specific policies, schemes, and commitments. UPSC tests whether you know what India has actually promised (NDC), what instruments exist to meet those promises (Carbon Market, Green Credit), and what the updated targets are. This section covers the seven most important developments from the last 18 months.

MUST-KNOW

2.1 India's Nationally Determined Contributions (NDCs) — the Panchamrit commitments

Prelims & Mains · Environment + Governance · Also: State PCS, CDS

India's current NDC under the Paris Agreement was updated in August 2022 (before COP27). Its core quantitative targets are to reduce emissions intensity of GDP by 45% from 2005 levels by 2030, and to achieve about 50% of cumulative electric power installed capacity from non-fossil-fuel-based sources by 2030. These are legally notified under the UNFCCC.

At COP26 Glasgow (2021), PM Modi announced the Panchamrit — five commitments. Two were enhanced in the 2022 NDC update. India's longer-term target is net-zero emissions by 2070. The third NDC (covering post-2030 targets) is expected to be submitted before COP31 (Antalya, 2026).

KEY FACTS FOR PRELIMS

Framework	Paris Agreement (2015) — Article 4 requires NDCs updated every 5 years
Current NDC	Updated August 2022 (before COP27 Sharm el-Sheikh)
Emissions intensity	Reduce by 45% below 2005 levels by 2030 (was 33–35% in original 2015 NDC)
Non-fossil capacity	~50% of cumulative installed electric power by 2030 (was 40%)
Carbon sink	Create additional 2.5–3 billion tonnes CO2 equivalent carbon sink by 2030 via forest & tree cover
Net-zero target	2070 (announced at COP26 Glasgow, 2021)
Panchamrit (COP26)	500 GW non-fossil capacity by 2030; 50% energy from renewables by 2030; 1 billion tonnes CO_2 equivalent cumulative emission reduction by 2030 (below projected levels); 45% emissions-intensity cut by 2030; Net-zero by 2070
Progress (2024)	India's carbon stock from forests at 30.43 billion tonnes CO2e — has already added ~2.29 Bt of the 2.5–3 Bt sink target

REVISE: *Shankar IAS / PMF IAS — Climate Change chapter + Paris Agreement. Keep a one-page sheet with Panchamrit + two enhanced NDC targets.*

PYQ LINK: *UPSC 2022 Prelims asked about India's INDC targets. UPSC 2023 Mains asked about India's climate commitments. Emissions-intensity and non-fossil-fuel percentage are the two numbers UPSC returns to.*

MUST-KNOW

2.2 Carbon Credit Trading Scheme (CCTS) — India's compliance carbon market

Prelims & Mains · Economy + Environment · Also: SSC, Banking

The CCTS was notified in June 2023 under the Energy Conservation (Amendment) Act, 2022. It creates India's first compliance carbon market — a legally binding, baseline-and-credit emissions trading system (ETS). In October 2025 and January 2026, the Ministry of Environment, Forest and Climate Change notified legally binding greenhouse gas emission intensity (GEI) targets for seven of the nine energy-intensive sectors initially covered.

CCTS is India's transition from the older PAT scheme (Perform, Achieve and Trade — an energy-efficiency scheme run by BEE since 2012). Entities that outperform their target earn Carbon Credit Certificates (CCCs); entities that fall short must buy and surrender CCCs. The first CCC trading on power exchanges is expected by mid-2026. On 21 March 2026, the Indian Carbon Market Portal was launched at the Prakriti 2026 conference.

KEY FACTS FOR PRELIMS

Legal basis	Energy Conservation (Amendment) Act, 2022 + CCTS notification, June 2023
Administrator	Bureau of Energy Efficiency (BEE), under Ministry of Power
Regulator	Central Electricity Regulatory Commission (CERC) — market oversight
Registry	Grid Controller of India Limited
National oversight	National Steering Committee for the Indian Carbon Market (NSCICM) — co-chaired by Secretary MoP and Secretary MoEFCC
Structure	Two mechanisms: (1) Compliance mechanism for obligated entities (baseline-and-credit); (2) Voluntary offset mechanism for non-obligated entities
Sectors (9)	Aluminium, Cement, Chlor-alkali, Pulp & Paper, Iron & Steel, Fertilizer, Petroleum Refining, Petrochemicals, Textiles
Baseline year	FY 2023–24 · Compliance years: 2025–26 and 2026–27
Gases covered	Scope 1 + Scope 2 + limited Scope 3 (CO2 and PFCs initially)
Penalty	Environmental compensation = 2× average market price of shortfall
Eight approved offset methodologies (March 2025)	Renewable energy (incl. hydro & pumped storage); green hydrogen; industrial energy efficiency; landfill methane recovery; mangrove afforestation/reforestation; RE with storage; offshore wind; compressed biogas

REVISE: *Shankar IAS / PMF IAS — Climate Change chapter. Link with Perform, Achieve & Trade (PAT) scheme (2012) run by BEE. Contrast with EU ETS (cap-and-trade) vs Indian CCTS (baseline-and-credit, intensity-based).*

PYQ LINK: *UPSC 2021 Prelims asked about carbon markets. UPSC 2023 asked about carbon pricing. CCTS is the current operational instrument — testable.*

SHOULD-KNOW

2.3 Green Credit Programme (GCP) — the 2025 revision

Prelims · Environment · Also: State PCS

The Green Credit Programme was notified on 12 October 2023 under the Environment (Protection) Act, 1986, as a voluntary market-based mechanism to incentivize environment-positive actions. Its first methodology (February 2024) focused on tree plantations on degraded forest lands — one credit per tree surviving, with density of 1,100 trees per hectare. The programme

attracted significant ecological criticism — for ignoring open natural ecosystems (ONEs), for double-counting with CCTS, and for over-dense plantation targets.

In August–September 2025, MoEFCC issued revised rules. The key shifts: credits are awarded after a five-year maintenance period based on canopy density of at least 40%, not tree count. Green Credits are now non-tradable and non-transferable (except between a holding and subsidiary company) and can be exchanged only once — for compensatory afforestation, CSR compliance, or project plantation obligations. Once exchanged, they are extinguished.

KEY FACTS FOR PRELIMS

Legal basis	Environment (Protection) Act, 1986 — Green Credit Rules, 2023
Announced at	G20 Summit 2023 by Prime Minister
Implementing agency	Indian Council of Forestry Research and Education (ICFRE)
2024 methodology	1 credit per tree, 1,100 trees/ha, minimum 2-year growth
2025 revision	5-year restoration period; 40% canopy density; non-tradable; one-time exchange only
Permitted uses	Compensatory afforestation (Van Sanrakshan Evam Samvardhan Adhiniyam, 1980); CSR; project plantation obligations
Planned activities	Tree plantation; water conservation; sustainable agriculture; waste management; air pollution reduction; mangrove conservation & restoration; ecomark label; sustainable building infrastructure
Key criticism	Still tree-centric — ignores open natural ecosystems (ONEs) like grasslands where afforestation harms water tables

REVISE: *Shankar IAS / PMF IAS — Environmental Policy chapter. Distinguish GCP (voluntary, tree-based) from CCTS (compliance, carbon-based).*

PYQ LINK: *UPSC frequently tests which ministry administers which scheme, and who are the target entities. GCP's non-tradable nature and the two-programme distinction (GCP vs CCTS) is classic Statement I/II material.*

SHOULD-KNOW

2.4 Mission LiFE — Lifestyle for Environment

Prelims · Environment · Also: State PCS

Mission LiFE was launched by Prime Minister Modi at COP27 (Sharm el-Sheikh, November 2022). It is a global initiative positioning individual and community behavioural change as a pillar of climate action, alongside technology and finance. The idea was first shared at COP26 Glasgow (2021). LiFE's five-year strategy (2022–2027) targets one billion Indians and foreign nationals as practitioners.

LiFE was formally adopted into UN climate processes at COP28 Dubai. At Ramsar COP15 (Zimbabwe, 2025), India introduced and passed a resolution explicitly linking Mission LiFE to wetland conservation — the first UN environmental convention to adopt a LiFE-linked resolution.

KEY FACTS FOR PRELIMS

Launch	COP27, Sharm el-Sheikh, November 2022 (concept at COP26, 2021)
Core idea	Lifestyle For Environment — Pro Planet People (P3) philosophy; 'mindful and deliberate utilization'

Seven themes	Save water; save energy; reduce waste; reduce e-waste; single-use plastic; sustainable food; healthy lifestyle
75 individual actions	Targeted across the 7 themes — curated by NITI Aayog
Timeline	Mission period 2022–2028
Target	1 billion LiFE practitioners (Indians and global citizens) by 2027
International	Adopted at COP28 Dubai UAE Consensus; Ramsar COP15 adopted LiFE-linked resolution (2025)
Scientific basis	UNEP & IEA estimates show demand-side measures can cut emissions by ~40–70% by 2050

REVISE: *Shankar IAS / PMF IAS — Climate Change chapter. Keep a two-line note linking LiFE to India's climate soft power pitch.*
PYQ LINK: *UPSC 2023 Prelims asked about LiFE movement. Expect more granular questions on which themes it covers.*

MUST-KNOW

2.5 National Green Hydrogen Mission — progress 2024–2026

Prelims & Mains · Energy + Environment · Also: State PCS, CDS

The National Green Hydrogen Mission was approved by the Union Cabinet in January 2023 with an outlay of Rs 19,744 crore (FY2023–24 to FY2029–30). It targets production of at least 5 million metric tonnes (MMT) of green hydrogen per annum by 2030, 125 GW of associated renewable energy capacity, and abatement of nearly 50 MMT of CO2 annually. Green hydrogen here means hydrogen produced by electrolysis of water using renewable energy.

The SIGHT programme (Strategic Interventions for Green Hydrogen Transition) is the mission's main financial lever — with two components: incentives for electrolyzer manufacturing, and incentives for green hydrogen production. Through 2024–25, over 862,000 tonnes/year of green hydrogen production capacity was allotted. The first indigenous 10 MW electrolyzer was commissioned. In 2025, eight sectors' carbon credit methodologies included green hydrogen production.

KEY FACTS FOR PRELIMS

Launched	January 2023 (Cabinet approval) · Outlay Rs 19,744 crore
Nodal Ministry	Ministry of New and Renewable Energy (MNRE)
2030 targets	5 MMT/year green hydrogen production · 125 GW renewable capacity addition · 50 MMT CO2 abated/year
Key programme	SIGHT — incentives for electrolyzer manufacturing + green hydrogen production
Pilot sectors	Steel, mobility, shipping, transportation, bunkering hubs
Green H2 definition	Emission intensity threshold ≤ 2 kg CO2 equivalent per kg of H2 (as per notification, August 2023)
Applications	Refineries (replacing grey H2), fertilizer (ammonia), steelmaking (DRI route), long-haul transport (fuel cells)

REVISE: *Shankar IAS / PMF IAS — Renewable Energy chapter. Link with PAT, CCTS, and hydrogen colour conventions (grey, blue, green, pink/red, turquoise).*
PYQ LINK: *UPSC 2023 Prelims asked about green hydrogen production. Colour classification of hydrogen is testable — as is the electrolysis basis for 'green.'*

SHOULD-KNOW

2.6 Sovereign Green Bonds — India's climate financing tool

Prelims & Mains · Economy + Environment · Also: SSC, Banking

Sovereign Green Bonds (SGrBs) are government debt securities whose proceeds are earmarked for green projects. India's first SGrB issuances began in January 2023 (Rs 16,000 crore raised across two tranches in FY2022–23). Since then, auctions have continued each half-year. As of FY2024–25, cumulative issuance exceeds Rs 40,000 crore. Proceeds fund projects in renewable energy, clean transport, climate adaptation, and sustainable water management.

India has a Sovereign Green Bonds Framework (November 2022) setting out eligible sectors and exclusions (no nuclear, no fossil fuels even for transition). A Green Finance Working Committee (GFWC) under the Ministry of Finance oversees allocation. RBI auctions SGrBs on behalf of the government, with participation from Foreign Portfolio Investors via the Fully Accessible Route (FAR).

KEY FACTS FOR PRELIMS

First issuance	January 2023 (Rs 8,000 crore tranche 1); Rs 16,000 crore in FY2022–23 total
Framework	Sovereign Green Bonds Framework, November 2022
Eligible sectors	Renewable energy; clean transportation; energy efficiency; climate adaptation; sustainable water & waste management; pollution prevention; terrestrial/aquatic biodiversity
Excluded	Fossil fuels (incl. 'transition' fuels); nuclear; large hydro with high environmental footprint; biomass from peatlands
Auction agent	Reserve Bank of India (RBI)
FPI access	Via Fully Accessible Route (FAR)
Tenors	5-year, 10-year, 30-year green bonds issued to date

REVISE: *Indian Economy by Ramesh Singh — Chapter on Financial Markets + Budget. NCERT Economy (Class XI) for bond basics.*
PYQ LINK: *UPSC 2023 Prelims asked about green bonds. Expect questions on India's SGrB framework — which sectors are / are not eligible.*

SHOULD-KNOW

2.7 Global Stocktake, CBDR-RC, and India's Negotiating Position

Prelims & Mains · Environment + International Affairs · Also: CDS

The Global Stocktake (GST) is a Paris Agreement process under Article 14 to assess collective progress every five years. The first GST concluded at COP28 Dubai (December 2023) with the UAE Consensus. Its key language — 'transitioning away from fossil fuels in energy systems' — was the first UN climate decision to reference fossil fuels explicitly. The UAE Consensus also called for tripling renewable energy capacity globally by 2030 and doubling energy efficiency improvement rates.

India has consistently anchored its negotiating position on CBDR-RC — the principle that developed and developing countries have 'common but differentiated responsibilities and respective capabilities' (UNFCCC Article 3.1). India argues that historical emissions, current per capita emissions, and development needs must shape differentiated action. India

rejected the COP29 NCQG figure of USD 300 billion as inadequate and opposes binding upstream caps on fossil fuel production.

KEY FACTS FOR PRELIMS

First GST outcome	COP28 Dubai UAE Consensus (Dec 2023)
Key phrase	'Transitioning away from fossil fuels in energy systems' — first UN climate text to name fossil fuels
Numerical goals	Triple renewables capacity globally by 2030; double rate of energy efficiency improvement
CBDR-RC	Common But Differentiated Responsibilities and Respective Capabilities (UNFCCC Art. 3.1)
India's per-capita CO2	~2 tonnes — roughly one-third of global average
Key asks by India	Grant-based finance (not loans); equitable carbon space; technology transfer without IPR barriers; no trade-restrictive unilateral measures
Concerns on CBAM	India opposes EU Carbon Border Adjustment Mechanism (CBAM) as a disguised trade barrier violating CBDR-RC

REVISE: *Shankar IAS / PMF IAS — Climate Change chapter + Paris Agreement. NCERT Economy (Class XII) for trade concepts like non-tariff barriers.*

PYQ LINK: *UPSC Mains 2017 asked about CBDR. UPSC 2023 Mains asked about India's climate stance. CBAM is likely a 2026 Prelims theme.*

SECTION 2 — PRACTICE

Test yourself

Q1. With reference to India's updated Nationally Determined Contribution (NDC) submitted in 2022, which of the following are correct?

(a) Reduce emissions intensity of GDP by 45% from 2005 levels by 2030
(b) Achieve about 50% of cumulative electric power installed capacity from non-fossil-fuel-based sources by 2030
(c) Create an additional carbon sink of 2.5 to 3 billion tonnes of CO2 equivalent by 2030
(d) Achieve net-zero emissions by 2050

Select the correct answer using the code given below:

(a) 1, 2 and 3 only
(b) 2, 3 and 4 only
(c) 1 and 4 only
(d) All four

Q2. The Carbon Credit Trading Scheme (CCTS), notified in 2023, has the following features. Choose the correct statements:

(a) It is administered by the Bureau of Energy Efficiency under the Ministry of Power.
(b) It is a cap-and-trade system with absolute emission caps for obligated entities.
(c) The Grid Controller of India Limited functions as the Registry.
(d) It was established under the Energy Conservation Act.

Select the correct answer using the code given below:

(a) 1, 3 and 4 only
(b) 1, 2 and 4 only
(c) 2 and 3 only
(d) All four

Q3. Match the following climate finance instruments with their correct descriptions:

(a) Sovereign Green Bonds — government debt securities; proceeds for green projects; fossil fuels excluded
(b) Carbon Credit Certificate — issued under CCTS compliance mechanism; tradable on power exchanges
(c) Green Credit — non-tradable; earned primarily for tree plantation on degraded forest land
(d) International Transfer Mitigation Outcome (ITMO) — bilateral carbon trade under Paris Agreement Article 6.2

How many of the above pairs are correctly matched?

(a) Only two
(b) Only three
(c) All four
(d) Only one

Q4. Statement I: India has excluded nuclear power projects from the list of sectors eligible for proceeds of Sovereign Green Bonds. Statement II: Sovereign Green Bonds are issued under a framework that aligns with international green bond principles requiring exclusion of non-renewable sources.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Q5. With reference to the National Green Hydrogen Mission, consider the following statements:

(a) It is implemented by the Ministry of New and Renewable Energy.
(b) Its 2030 target is to produce at least 5 million metric tonnes of green hydrogen per year.
(c) Green hydrogen under the mission must be produced using fossil fuels with carbon capture and storage.
(d) The SIGHT programme provides financial incentives for electrolyzer manufacturing and green hydrogen production.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1, 3 and 4 only
(c) 2, 3 and 4 only
(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (a) 1, 2 and 3 only

Statement 4 is incorrect — India's net-zero target is 2070, not 2050 (announced at COP26 Glasgow under Panchamrit). The 45% emissions-intensity cut, 50% non-fossil capacity, and 2.5–3 Bt additional carbon sink are all part of the updated 2022 NDC.

Q2. Answer: (a) 1, 3 and 4 only

Statement 2 is incorrect — CCTS is a baseline-and-credit, intensity-based system, not a cap-and-trade system with absolute caps like the EU ETS. Entities have GHG emission intensity targets (tCO_2e per unit of output) and earn CCCs for outperforming their targets.

Q3. Answer: (c) All four

All four pairs are correctly matched. SGrBs exclude fossil fuels and nuclear. CCCs are tradable on power exchanges under CERC regulation. Green Credits (post-2025 revision) are non-tradable except between holding and subsidiary. ITMOs are the Article 6.2 instrument for bilateral transfers of mitigation outcomes.

Q4. Answer: (c)

Statement I is correct — India's Sovereign Green Bonds Framework (Nov 2022) specifically excludes nuclear projects. Statement II is incorrect — international green bond principles (such as ICMA's Green Bond Principles) do not require the exclusion of nuclear; India's exclusion is a domestic policy choice, not an international mandate. Many countries include nuclear in their green taxonomies (notably the EU).

Q5. Answer: (a) 1, 2 and 4 only

Statement 3 is incorrect — 'Green' hydrogen specifically means hydrogen produced via electrolysis of water using renewable energy, not fossil fuels with CCS (that would be 'blue' hydrogen). The MNRE notification defines green hydrogen by an emissions threshold of ≤ 2 kg CO_2e per kg H_2.

SECTION 03

Biodiversity & Species in News

IUCN updates, new species, and India's conservation flagships

Biodiversity has appeared in every UPSC Prelims paper from 2011 to 2025 without exception — it is the single most reliable question-block in the environment section. UPSC tests three things: IUCN Red List categories and what species fall where, species-specific facts (habitat, endemism, threats), and flagship conservation projects. The nine topics here cover everything that has moved in the last 18 months.

MUST-KNOW

3.1 IUCN Red List — 2024 and 2025 updates for Indian species

Prelims · Environment · Also: State PCS, CDS

The IUCN Red List is updated at least twice a year, timed with major conservation events. Two updates matter most for 2026 Prelims: the October 2024 update, and the October 2025 update released at the IUCN World Conservation Congress in Abu Dhabi (9–15 October 2025) — informally called 'the Olympics of Conservation.'

In October 2025, the Red List update modified the conservation status of twelve Indian bird species. Eight species were downlisted (positive trend); four were uplisted. The four uplisted species all depend on open natural ecosystems — grasslands, semi-arid landscapes, scrublands — and signal the quiet collapse of these habitats. Long-billed Grasshopper-warbler was uplisted to Endangered.

KEY FACTS FOR PRELIMS

Nine IUCN categories	EX → EW → CR → EN → VU → NT → LC → DD → NE
CR criteria	>80% decline in 10 years OR <50 mature individuals OR 50% extinction probability in 10 years
India CR additions (2024)	8 animal species + 7 plant species added as CR; 1 plant (Limnophila limnophiloides) listed Extinct
India CR additions (2025)	13 more animal + 13 plant species added as CR (as of 21 October 2025)
Birds downlisted (Oct 2025)	8 species — positive trend; conservation working
Birds uplisted (Oct 2025)	4 species; Long-billed Grasshopper-warbler → EN; all depend on Open Natural Ecosystems
BirdLife International	Red List Authority for birds under IUCN; also identifies Important Bird & Biodiversity Areas (IBAs)
Who coined 'biodiversity hotspot'	Norman Myers (not BirdLife) — common UPSC trap

REVISE: *Shankar IAS / PMF IAS — Biodiversity chapter. Memorize nine IUCN categories in order + criteria for CR and EN.*

PYQ LINK: *UPSC 2019, 2020, 2022 tested IUCN Red List categories. Most reliable 2-mark block in the paper.*

MUST-KNOW

3.2 India's first National Red List Assessment Initiative (2025–2030)

Prelims & Mains · Environment · Also: State PCS

India unveiled its first National Red List of Endangered Species at the IUCN World Conservation Congress in Abu Dhabi on 9–15 October 2025. The five-year National Red List Assessment Initiative (2025–2030) will systematically evaluate the extinction risk of approximately 11,000 Indian species — 7,000 flora and 4,000 fauna — using IUCN standards.

Before this initiative, only about 7.2% of Indian fauna and 6.33% of flora had been globally assessed by IUCN — a huge data gap. The initiative fulfills Target 4 of the Kunming-Montreal Global Biodiversity Framework.

KEY FACTS FOR PRELIMS

Launched at	IUCN World Conservation Congress, Abu Dhabi, 9–15 October 2025
Duration	2025–2030 (five-year initiative)
Species to be assessed	~11,000 (7,000 flora + 4,000 fauna)
Nodal Ministry	Ministry of Environment, Forest and Climate Change (MoEFCC)
Implementing bodies	Botanical Survey of India (BSI), Zoological Survey of India (ZSI), Wildlife Institute of India (WII)
Global link	Fulfills Target 4 of Kunming-Montreal Global Biodiversity Framework
Prior assessment gap	Only 7.2% of fauna and 6.33% of flora globally assessed before

REVISE: *Shankar IAS / PMF IAS — Biodiversity + Indian institutions (BSI, ZSI, WII). Know each agency's mandate.*
PYQ LINK: *UPSC 2015 asked about WII. UPSC 2013 asked about BSI. Institutional mandate questions are recurring.*

MUST-KNOW

3.3 Project Cheetah — from Kuno to Gandhi Sagar and beyond

Prelims & Mains · Environment · Also: State PCS, CDS

Project Cheetah, launched on 17 September 2022 at Kuno National Park (Madhya Pradesh), is the world's first intercontinental large-carnivore translocation project. Cheetahs — extinct in India since 1952 — have been brought from Namibia (2022), South Africa (2023), and Botswana (February 2026). Hosted under the National Tiger Conservation Authority (NTCA).

In March 2025, cheetahs were relocated to Gandhi Sagar Wildlife Sanctuary in northern Madhya Pradesh as a second site. As of 2026, India's cheetah population is 57 — a milestone reached when KGP-2, an Indian-born female (daughter of Gamini from Tswalu), delivered four cubs in the wild. It was the first wild litter born to an Indian-born female.

KEY FACTS FOR PRELIMS

Launch	17 September 2022, Kuno National Park, Madhya Pradesh
Implementing body	National Tiger Conservation Authority (NTCA) — statutory under Wildlife Protection Act 1972
Source countries	Namibia (2022), South Africa (2023), Botswana (February 2026)
Species imported	African Cheetah — NOT the Asiatic Cheetah (CR, survives only in Iran)

2025 milestone	Relocation to Gandhi Sagar WLS, northern MP — March 2025
Gandhi Sagar WLS	Near MP-Rajasthan border; Khathiar-Gir dry deciduous ecoregion; Chambal River flows through
2026 milestone	KGP-2 (Indian-born) birthed first wild litter; population reached 57
Target	Metapopulation of 60–70 cheetahs across multiple landscapes by 2032
Proposed corridor	Kuno–Gandhi Sagar: ~17,000 sq km across 8 MP + 7 Rajasthan districts

REVISE: *Shankar IAS / PMF IAS — Wildlife Conservation + Project Tiger + Project Elephant. NCERT Class XII Biology for extinction concepts.*

PYQ LINK: *UPSC 2024 Prelims referenced cheetah reintroduction. Expect explicit questions on source countries, NTCA as statutory body, and African vs Asiatic distinction.*

SHOULD-KNOW

3.4 Invasive Alien Species & the IPBES report

Prelims · Environment · Also: State PCS

Invasive alien species (IAS) are non-native species that become established outside their natural range and cause ecological, economic, or health damage. The IPBES Invasive Alien Species Assessment (2023) found over 37,000 alien species established globally; 3,500+ are invasive and cause USD 423 billion in annual losses. In India, 66 invasive species are currently recognized as 'major'.

Three Indian invasives are repeatedly in news: Lantana camara (woody shrub overrunning ~40% of tiger habitat); Prosopis juliflora (vilayati kikar, dominating drylands of Rajasthan and Gujarat); and Water hyacinth (Eichhornia crassipes, clogging Loktak and Dal Lake). The IUCN Invasive Species Specialist Group (ISSG) was tested in UPSC 2023 Prelims.

KEY FACTS FOR PRELIMS

IPBES report	Invasive Alien Species Assessment 2023 — 37,000 alien species; 3,500 invasive
Global losses	USD 423 billion annually
IUCN body	Invasive Species Specialist Group (ISSG) under IUCN Species Survival Commission
Three major Indian invasives	Lantana camara (forests); Prosopis juliflora (drylands); Eichhornia crassipes (wetlands)
Loktak (Manipur)	Heavily affected by Eichhornia; on Montreux Record
Lantana spread	~40% of India's tiger habitat overrun
Legal tool	Wildlife Protection Act 1972 Section 37 empowers Central Govt to notify invasive species

REVISE: *Shankar IAS / PMF IAS — Biodiversity threats chapter. Note the three main Indian invasives + their ecosystems.*

PYQ LINK: *UPSC 2023 asked about IUCN Invasive Species Specialist Group. UPSC 2012 asked about Lantana.*

SHOULD-KNOW

3.5 New and rediscovered species in India (2024–2026)

Prelims · Environment · Also: State PCS

India continues to be a global biodiversity frontier. The Zoological Survey of India's Animal Discoveries reports document 600+ new animal species and subspecies per year. Notable discoveries: Lycodon irwini (a new wolf snake on Great Nicobar, described 2025); Chel Snakehead (Channa amphibia) rediscovered after 80 years in the Chel River, West Bengal (2024).

In October 2025, researchers documented undescribed species on Great Nicobar — including the yet-to-be-formally-named 'Great Nicobar crake' — as part of the EIA debate around the Great Nicobar development project. These discoveries add weight to scientific concerns about the project.

KEY FACTS FOR PRELIMS

ZSI Animal Discoveries	Annual publication; 600+ new animal species/records each year
BSI Plant Discoveries	Annual publication; hundreds of new plant species/records
New species 2025	Lycodon irwini (Great Nicobar wolf snake)
Rediscovered (2024)	Chel Snakehead (Channa amphibia) — Chel River, West Bengal; last seen 80 years ago
Great Nicobar endemics	Undescribed taxa including 'Great Nicobar crake' (bird, not yet formally named)
Four Indian biodiversity hotspots	Western Ghats-Sri Lanka; Indo-Burma; Himalaya; Sundaland (Nicobar Islands)

REVISE: *Shankar IAS / PMF IAS — Biodiversity Hotspots chapter. Four hotspots partially in India.*
PYQ LINK: *UPSC 2015 asked which is not a biodiversity hotspot. UPSC 2018 asked about Western Ghats.*

MUST-KNOW

3.6 Flagship species population updates — tigers, lions, elephants, snow leopards

Prelims · Environment · Also: State PCS, CDS

India hosts the majority of the world's wild populations of four flagship species: Bengal tigers (~75%), Asiatic lions (100%), Asian elephants (~60%), and a significant share of snow leopards. Each has a dedicated conservation programme.

Tiger Census 2022 put India at 3,682 wild tigers — up from 2,967 in 2018 and 1,411 in 2006 (the Sariska shock). Madhya Pradesh leads with 785; Jim Corbett holds the most tigers in any single reserve (231). Snow Leopard Population Assessment (SPAI), the first-ever scientific estimate, was released in January 2024 and put India's snow leopard population at 718. The 2025 Asiatic Lion census put the population at ~891.

KEY FACTS FOR PRELIMS

Bengal Tiger (IUCN EN)	3,682 (2022); 74–75% of world wild tigers; MP 785, KA 563, UK 560
Project Tiger	Launched 1 April 1973 by PM Indira Gandhi; under NTCA (statutory body since 2006)
Asiatic Lion (IUCN EN)	~891 (2025 census); endemic to Gir NP & WLS, Gujarat; Project Lion (2020)
Asian Elephant (IUCN EN)	~27,000–29,000 in India; Project Elephant (1992); 33 Elephant Reserves

Snow Leopard (IUCN VU)	718 in India (SPAI 2024 — first scientific estimate)
SPAI range states	Ladakh, J&K, HP, Uttarakhand, Sikkim, Arunachal Pradesh
One-Horned Rhino (IUCN VU)	~4,000 in Kaziranga (Assam); Project Rhino
Gangetic Dolphin (IUCN EN)	National Aquatic Animal (2009); Project Dolphin (2020)

REVISE: *Shankar IAS / PMF IAS — Wildlife Conservation. Keep a table: species → IUCN category → population → habitat → flagship project.*

PYQ LINK: *UPSC 2020 asked about Project Tiger. UPSC 2017 asked about Gir. Species-project-habitat triplets are tested almost every year.*

SHOULD-KNOW

3.7 Pangolin, Great Indian Bustard, Dugong — species in continuous news

Prelims · Environment · Also: State PCS

Three species have featured repeatedly in environmental news. Indian Pangolin (IUCN CR) is the world's most trafficked wild mammal — scales used in traditional medicine. Schedule I of Wildlife Protection Act gives it highest protection.

Great Indian Bustard (IUCN CR, <150 left) was central to a Supreme Court case (March 2024) that modified earlier orders mandating undergrounding of power lines across vast areas of Rajasthan and Gujarat. An expert committee was constituted to balance renewable energy transmission with bustard conservation. Dugong — the only herbivorous marine mammal in India — is protected via India's first Dugong Conservation Reserve in Palk Bay, Tamil Nadu (notified September 2022).

KEY FACTS FOR PRELIMS

Indian Pangolin (CR)	Most trafficked mammal; scales = keratin; Schedule I WLPA; peninsular India
Great Indian Bustard (CR)	<150 left; Ardeotis nigriceps; Thar Desert (Rajasthan), Gujarat, parts of MP
GIB Supreme Court case	March 2024 — SC modified 2021 undergrounding order; expert committee formed
Project GIB	Conservation breeding at Sam, Jaisalmer; partnership with WII
Dugong (IUCN VU)	Only herbivorous marine mammal in India; Schedule I WLPA
Dugong Conservation Reserve	India's first — Palk Bay, Tamil Nadu (notified 21 Sept 2022); ~448 sq km
Other species in news	Hangul/Kashmiri Stag (CR, Dachigam NP); Pygmy Hog (CR, smallest pig, Assam); Nilgiri Tahr (TN state animal; Project Nilgiri Tahr 2023)

REVISE: *Shankar IAS / PMF IAS — Wildlife Protection Act 1972 + Schedules. The 2022 amendment reorganized Schedules.*

PYQ LINK: *UPSC 2020 asked about GIB. UPSC 2022 asked about dugong. These are UPSC favourites — each appears every 2–3 years.*

SHOULD-KNOW

3.8 Wildlife Protection (Amendment) Act 2022 — implementation status

Prelims · Environment + Polity · Also: State PCS

The Wild Life (Protection) Amendment Act 2022 came into force on 1 April 2023. It significantly revised the Schedules (earlier six, now four): Schedule I (highest protection), Schedule II (other protected animals), Schedule III (protected plants),

Schedule IV (CITES-listed species). The Act also implements India's obligations under CITES (1973 — India a party since 1976).

Important changes: Management Plans are now mandatory for sanctuaries and national parks; elephant transport/transfer is allowed only for bona fide religious/cultural purposes with written permission of the Chief Wildlife Warden; penalties have been stiffened significantly.

KEY FACTS FOR PRELIMS

Act title	Wild Life (Protection) Amendment Act, 2022 — effective 1 April 2023
Schedules revised	Earlier 6 → now 4 (Schedule I highest; Schedule IV = CITES species)
CITES link	Act implements India's CITES obligations; CITES has 3 Appendices (I, II, III)
CITES Appendix I	Threatened with extinction — international trade prohibited
CITES Appendix II	Trade allowed with permit
CITES Appendix III	Species listed unilaterally by a single country
India's protected areas	106 National Parks + 573 Wildlife Sanctuaries + 122 Conservation Reserves + 14 Community Reserves + 18 Biosphere Reserves

REVISE: *Shankar IAS / PMF IAS — Wildlife Conservation + Environmental Laws. Keep a comparison: WLPA Schedules vs CITES Appendices.*

PYQ LINK: *UPSC regularly tests WLPA Schedules. The 2022 amendment is new — expect direct testing.*

MUST-KNOW

3.9 Kunming-Montreal Global Biodiversity Framework (KMGBF) — the 30×30 agenda

Prelims & Mains · Environment + International Affairs · Also: State PCS

Adopted at CBD COP15 (Montreal, December 2022), the Kunming-Montreal Global Biodiversity Framework (KMGBF) is the global biodiversity master plan — the equivalent of what the Paris Agreement is for climate. It has four long-term goals (for 2050) and 23 action-oriented targets (for 2030).

The headline target is Target 3 — the '30 by 30' goal: conserve and manage at least 30% of the world's terrestrial, inland water, coastal, and marine areas through protected areas and Other Effective Area-based Conservation Measures (OECMs) by 2030. Target 2 calls for effective restoration of at least 30% of degraded ecosystems. India updated its National Biodiversity Strategy and Action Plan (NBSAP) in 2024 to align with KMGBF.

KEY FACTS FOR PRELIMS

Adopted at	CBD COP15, Montreal, 19 December 2022
Structure	4 long-term goals (for 2050) + 23 targets (for 2030)
Target 3 (30×30)	Conserve ≥30% of land, inland water, coastal & marine areas by 2030
Target 2	Restore at least 30% of degraded ecosystems
Target 4	Halt human-induced extinctions — basis for India's National Red List Initiative
Target 18	Eliminate harmful subsidies worth USD 500 billion/year by 2030

Target 19	Mobilize USD 200 billion/year; developed countries to provide USD 20 bn/yr by 2025, USD 30 bn/yr by 2030
Funding vehicle	Global Biodiversity Framework Fund (GBFF) — hosted by GEF, launched August 2023
India's NBSAP	Updated in 2024 to align with KMGBF

REVISE: *Shankar IAS / PMF IAS — Biodiversity + CBD chapter. Memory hook: KMGBF = 30×30 + 23 targets + GBFF.*
PYQ LINK: *KMGBF is the most likely Prelims question from biodiversity governance for 2026. UPSC 2022 asked about CBD targets indirectly.*

SECTION 3 — PRACTICE

Test yourself

Q1. With reference to Project Cheetah in India, consider the following statements:

(a) The project involves the reintroduction of the African Cheetah, not the Asiatic Cheetah.
(b) It is implemented under the National Tiger Conservation Authority (NTCA).
(c) Cheetahs have been brought from Namibia, South Africa, and Botswana.
(d) Gandhi Sagar Wildlife Sanctuary in Rajasthan is the second identified cheetah site.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1, 3 and 4 only
(c) 2, 3 and 4 only
(d) All four

Q2. The Kunming-Montreal Global Biodiversity Framework (2022) includes which of the following targets?

(a) Conserve at least 30% of the world's land and marine areas by 2030
(b) Restore at least 30% of degraded ecosystems by 2030
(c) Reduce global emissions by 43% below 2019 levels by 2030
(d) Mobilize USD 200 billion per year for biodiversity by 2030

Select the correct answer using the code given below:

(a) 1, 2 and 3 only
(b) 1, 2 and 4 only
(c) 2, 3 and 4 only
(d) 1, 3 and 4 only

Q3. Which of the following species is correctly matched with its IUCN Red List status and Indian habitat?

(a) Indian Pangolin — Critically Endangered — Peninsular India
(b) Great Indian Bustard — Endangered — Thar Desert
(c) Asiatic Lion — Vulnerable — Gir National Park
(d) Snow Leopard — Endangered — Western Ghats

Q4. Statement I: The IUCN Red List is updated at least twice a year, timed with major conservation events. Statement II: The IUCN World Conservation Congress is held every four years, with the 2025 edition held in Abu Dhabi, UAE.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I

(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I

(c) Statement I is correct but Statement II is incorrect

(d) Statement I is incorrect but Statement II is correct

Q5. Under the Wild Life (Protection) Amendment Act, 2022, which of the following is correct?

(a) The number of schedules has been reduced from six to four.

(b) Schedule IV now contains species listed under CITES Appendices.

(c) Transportation of live elephants is permitted for bona fide religious and cultural purposes with permission of the Chief Wildlife Warden.

(d) Management Plans are now mandatory for National Parks and Wildlife Sanctuaries.

Which of the statements given above are correct?

(a) 1 and 2 only

(b) 1, 2 and 3 only

(c) 1, 3 and 4 only

(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (a) 1, 2 and 3 only

Statement 4 is incorrect — Gandhi Sagar Wildlife Sanctuary is in Madhya Pradesh (north-western MP near the Rajasthan border), not in Rajasthan. Cheetahs were relocated there in March 2025 as the second site.

Q2. Answer: (b) 1, 2 and 4 only

Statement 3 is incorrect — the 43% emissions reduction target is a climate target (IPCC AR6 1.5°C pathway), not a KMGBF target. KMGBF is a biodiversity framework.

Q3. Answer: (a)

Only (a) is correct. Indian Pangolin is CR and found across peninsular India. GIB is CR (not EN); Asiatic Lion is EN (not VU); Snow Leopard is VU and lives in the Himalayas (not Western Ghats).

Q4. Answer: (b)

Both statements are correct. However, the quadrennial frequency of the IUCN Congress is not the explanation for the bi-annual frequency of Red List updates — updates coincide with many events, not just the Congress.

Q5. Answer: (d) All four

All four are correct. The 2022 amendment reorganized six schedules into four; Schedule IV mirrors CITES Appendices; elephant transport is now restricted; Management Plans mandatory.

SECTION 04

Protected Areas & Conservation

New Ramsar sites, Tiger Reserves, and the Great Nicobar debate

Protected areas are the ground on which India's biodiversity commitments stand. UPSC tests three recurring themes: sites recently added, distinctions between categories (NPs, Sanctuaries, Conservation Reserves, Biosphere Reserves, Tiger Reserves, Ramsar sites), and site-specific facts. This section covers the six most important developments from the last 18 months.

MUST-KNOW

4.1 Ramsar Sites — India now at 98, the highest in Asia

Prelims · Environment · Also: State PCS, CDS

India's Ramsar tally has grown from 75 in 2022 to 98 as of January 2026 — the highest in Asia and third-highest globally (after the UK and Mexico). Tamil Nadu leads with 20 Ramsar sites. Two sites — Keoladeo National Park (Rajasthan) and Loktak Lake (Manipur) — remain on the Montreux Record. Chilika Lake (Odisha) was on the Montreux Record from 1993 but removed in 2002 after successful restoration.

Additions in 2025–26 include: Khichan and Menar (Rajasthan, June 2025); Gokul Jalashay, Udaipur Jheel and Gogabeel Lake (Bihar); Khecheopalri Wetland (Sikkim — first Ramsar for the state); Udhwa Lake (Jharkhand — first for the state); Sakkarakottai & Therthangal Bird Sanctuaries (Tamil Nadu); Kopra Jalashay (Chhattisgarh — first for the state, December 2025); Siliserh Lake (Rajasthan, December 2025); Patna Bird Sanctuary (UP, 97th); Chhari-Dhand (Gujarat, 98th).

KEY FACTS FOR PRELIMS

Total Ramsar sites	98 (as of January 2026); highest in Asia, 3rd globally
India ratified	1 February 1982
First two sites (1981)	Chilika Lake (Odisha) + Keoladeo NP (Rajasthan)
Largest Ramsar site	Sundarban Wetland (West Bengal) — 4,23,000 hectares
Smallest Ramsar site	Renuka Wetland (Himachal Pradesh) — 20 hectares
State leader	Tamil Nadu — 20 Ramsar sites (highest)
Montreux Record (India)	Keoladeo NP, Rajasthan + Loktak Lake, Manipur
First-ever Ramsar in 2025	Chhattisgarh (Kopra Jalashay); Sikkim (Khecheopalri); Jharkhand (Udhwa)
97th & 98th	Patna Bird Sanctuary (UP); Chhari-Dhand (Gujarat)
Convention adopted	Ramsar, Iran, 1971; entered force 1975; World Wetlands Day 2 February

REVISE: *Shankar IAS / PMF IAS — Wetlands chapter. Know the Ramsar wetland definition (including marine water to 6 m depth).*

PYQ LINK: *UPSC 2019 asked about Ramsar sites. UPSC 2020 asked about Montreux Record. Recurring theme.*

MUST-KNOW

4.2 Tiger Reserves — 58 with Madhav, Ratapani, Guru Ghasidas additions

Prelims · Environment · Also: State PCS, CDS

India has 58 tiger reserves as of 2026, covering ~84,487 sq km (~2.56% of India's land area, per NTCA), under Project Tiger (launched 1 April 1973) managed by the National Tiger Conservation Authority (NTCA, statutory since 2006). Madhya Pradesh leads all states with 9 tiger reserves and 785 tigers.

Recent additions: Guru Ghasidas-Tamor Pingla (Chhattisgarh, 56th, 2024) — MP-Jharkhand corridor, third largest TR; Ratapani (MP, 57th, December 2024) — near Bhopal and Bhimbetka; Madhav (MP, 58th, March 2025) — Shivpuri district, Chambal region, 1,751 sq km, formerly royal hunting ground. Three reserves report zero tigers: Dampa (Mizoram), Buxa (West Bengal), Palamau (Jharkhand).

KEY FACTS FOR PRELIMS

Total Tiger Reserves	58 (March 2025); ~84,487 sq km (~2.56% of geographical area per NTCA)
Project Tiger	1 April 1973 by PM Indira Gandhi; 9 original reserves
NTCA	Statutory since 2006 (Wildlife Protection Act Section 38)
Tiger population 2022	3,682 (75% of world); up from 1,411 in 2006 (Sariska shock)
Largest TR	Nagarjunsagar-Srisailam (AP+Telangana, ~3,296 sq km)
Smallest TR	Bor (Maharashtra, 138 sq km)
First TR	Jim Corbett (Uttarakhand, 1973); most tigers in a reserve (231)
58th TR	Madhav, MP (10 March 2025); 1,751 sq km; Shivpuri
57th TR	Ratapani, MP (Dec 2024); 1,271 sq km
56th TR	Guru Ghasidas-Tamor Pingla, CG (2024)
Zero tigers	Dampa (MZ); Buxa (WB); Palamau (JH)

REVISE: *Shankar IAS / PMF IAS — Wildlife Conservation. Memorize top 3 tiger states (MP, KA, UK) and top 3 largest tiger reserves.*

PYQ LINK: *UPSC 2017 asked about NTCA. UPSC 2020 asked about Project Tiger. New tiger reserves are standard material.*

SHOULD-KNOW

4.3 Biosphere Reserves, Eco-Sensitive Zones, and Conservation Reserves

Prelims · Environment · Also: State PCS

India has 18 Biosphere Reserves, of which 12 are in UNESCO's World Network of Biosphere Reserves (WNBR). A Biosphere Reserve is a UNESCO designation (Man and the Biosphere Programme, 1971) with three zones — Core, Buffer, and Transition. They are larger than national parks and designed for integrating conservation with sustainable human use.

Eco-Sensitive Zones (ESZs) are buffer areas around protected areas, notified under the Environment Protection Act 1986. Typically extend up to 10 km from the boundary. Mining is prohibited; tourism and construction are regulated. In April 2025, the Supreme Court in T.N. Godavarman reaffirmed the 1 km default ESZ around every NP and WLS unless specifically notified otherwise.

KEY FACTS FOR PRELIMS

Biosphere Reserves	18 total; 12 in UNESCO's WNBR
First BR in India	Nilgiri Biosphere Reserve (1986)
Added to WNBR recently	Panna BR (MP) — added 2020
BR three zones	Core (strict protection) + Buffer (research) + Transition (sustainable use)
ESZ	Buffer around PAs under Environment Protection Act 1986; up to 10 km typical
SC T.N. Godavarman (April 2025)	Reaffirmed 1 km default ESZ around every PA unless notified
Conservation Reserves	122 total; state-owned land
Community Reserves	14 total; community-owned land

REVISE: *Shankar IAS / PMF IAS — Protected Area Network. Memorize the three BR zones and ESZ activity restrictions.*
PYQ LINK: *UPSC 2020 asked about biosphere reserves. The Panna addition and ESZ-related cases are testable.*

MUST-KNOW

4.4 Great Nicobar Development Project — biodiversity vs strategy debate

Prelims & Mains · Environment + Governance + Internal Security · Also: State PCS

The ₹75,000–81,000 crore 'Holistic Development of Great Nicobar Island' project (NITI Aayog conception) includes an International Container Transhipment Port (Galathea Bay), an international airport at Campbell Bay, a power plant, and a new township. Environmental clearance granted by MoEFCC in November 2022. Requires diversion of ~130 sq km of forest land and felling of ~8.5–9.6 lakh trees.

Key concerns: Galathea Bay is a primary nesting site for the Giant Leatherback Sea Turtle and the endemic Nicobar Megapode; the island is home to the Shompen (PVTG, ~200 individuals) and the Great Nicobarese; it lies in Seismic Zone V (highest). In October 2025 the NGT upheld the EC; on 30 October 2025 MoEFCC issued an order waiving litigation-stall time from EC validity. Galathea Bay WLS and Megapode WLS were denotified in January 2021.

KEY FACTS FOR PRELIMS

Project cost	Rs 75,000–81,000 crore (revised 2025); 30-year implementation
Components	Galathea transhipment port + Campbell Bay international airport + Township + Power plant
Forest diversion	~130 sq km; ~8.5–9.6 lakh trees
Ecological concerns	Leatherback Sea Turtle nesting; Nicobar Megapode endemic; Great Nicobar Biosphere Reserve
Tribal concerns	Shompen (PVTG, ~200) + Great Nicobarese; 92% of island is tribal reserve
Seismic risk	Zone V; 2004 tsunami sank southern tip by ~15 feet
NGT decision	October 2025 — upheld environmental clearance
MoEFCC order (30 Oct 2025)	Waives litigation-stall time from EC validity
Denotifications (Jan 2021)	Galathea Bay WLS and Megapode WLS denotified

REVISE: *Shankar IAS / PMF IAS — Coastal Ecosystems + Biosphere Reserves. NCERT for tribal communities; Forest Rights Act 2006.*
PYQ LINK: *Great Nicobar is a prime Prelims-Mains overlap topic for 2026.*

SHOULD-KNOW

4.5 Marine Protected Areas and Coral Reef Bleaching

Prelims · Environment · Also: State PCS

India has 132 Marine Protected Areas (MPAs). Four major MPA clusters protect coral reefs, mangroves, seagrass beds, and sea turtle nesting habitats: Gulf of Mannar (TN), Gulf of Kutch (Gujarat), Sundarbans (WB), and the Lakshadweep/Andaman archipelagos.

In April 2024, NOAA and the International Coral Reef Initiative confirmed the fourth global mass coral bleaching event — the second in 10 years — linked to record ocean temperatures. India's Lakshadweep and Andaman reefs suffered significant bleaching. The Global Coral Reef Monitoring Network estimates ~14% of coral reefs have been lost since 2009.

KEY FACTS FOR PRELIMS

MPAs in India	132 total
Four main coral areas	Gulf of Mannar + Gulf of Kutch + Lakshadweep + Andaman & Nicobar
Gulf of Mannar	India's first Marine Biosphere Reserve (1989); 21 islands; in UNESCO WNBR
Gulf of Kutch	Gujarat; India's first Marine National Park (1982)
Fourth global bleaching	Confirmed April 2024 by NOAA + ICRI
Coral types	Hermatypic (reef-building, with zooxanthellae) vs ahermatypic
Zooxanthellae	Symbiotic algae; expelled under thermal stress = bleaching

REVISE: *Shankar IAS / PMF IAS — Marine Ecosystems + Coral Reefs. NCERT Class XI Geography for Oceanography basics.*
PYQ LINK: *UPSC 2018 asked about coral reef distribution. Causes of bleaching are testable.*

SHOULD-KNOW

4.6 Western Ghats, Aravalli, and UNESCO World Heritage updates

Prelims · Environment + Geography · Also: State PCS

The Western Ghats — one of four global biodiversity hotspots partially in India — remain the centre of repeated debate. The Kasturirangan Committee (2013) recommended declaring 37% (~59,940 sq km) as Ecologically Sensitive Areas. MoEFCC has issued a series of draft notifications (sixth in July 2024) without final notification. ISFR 2023 noted for the first time that the Western Ghats ESA has lost 58.22 sq km of forest cover since 2013.

Aravalli Green Wall Project was launched in 2023 as a 1,400 km, 5 km wide green corridor across Haryana, Rajasthan, Gujarat, and Delhi to halt desertification. India has 43 UNESCO World Heritage Sites (as of early 2026). Recent additions include Moidams of the Ahom Dynasty (Assam, 2024) and Maratha Military Landscapes (2025 nomination). Khangchendzonga National Park (Sikkim) remains India's only Mixed Natural+Cultural WHS.

KEY FACTS FOR PRELIMS

Western Ghats hotspot	4 biodiversity hotspots partially in India; 6 states (GJ, MH, KA, GA, KL, TN)
Kasturirangan Committee	2013; recommended 37% (~59,940 sq km) as ESA
MoEFCC draft notifications	6 drafts; latest July 2024; final pending
ISFR 2023 finding	Western Ghats ESA lost 58.22 sq km forest cover since 2013
Aravalli Green Wall	2023; 1,400 km long, 5 km wide; HR + RJ + GJ + Delhi
UNESCO WHS (India)	43 total (early 2026)
Recent additions	Moidams of Ahom Dynasty (Assam, 2024 — 43rd)
Only Mixed Natural+Cultural WHS	Khangchendzonga NP (Sikkim)

REVISE: *Shankar IAS / PMF IAS — Biodiversity Hotspots. NCERT Class XI Geography for Western Ghats and Aravalli physiography.*

PYQ LINK: *UPSC 2018 asked about Western Ghats ESAs. UPSC 2013 asked about UNESCO WHS. New WHS nominations recur.*

SECTION 4 — PRACTICE

Test yourself

Q1. Consider the following statements regarding India's Ramsar sites:

(a) India ratified the Ramsar Convention in 1982 and currently has the highest number of Ramsar sites in Asia.
(b) Chilika Lake was the first Ramsar site in India and has been on the Montreux Record continuously since 1993.
(c) Tamil Nadu has the highest number of Ramsar sites among Indian states.
(d) Chhari-Dhand in Gujarat was added as India's 98th Ramsar site in early 2026.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1, 3 and 4 only
(c) 2 and 3 only
(d) All four

Q2. With reference to India's Tiger Reserves, consider the following:

(a) Project Tiger was launched in 1973 and the National Tiger Conservation Authority is a statutory body.
(b) Madhav Tiger Reserve in Madhya Pradesh was declared India's 58th Tiger Reserve in March 2025.
(c) Nagarjunsagar-Srisailam is India's largest Tiger Reserve and spans Andhra Pradesh and Telangana.
(d) The first Tiger Reserve of India was Palamau Tiger Reserve in Jharkhand.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1, 3 and 4 only
(c) 2, 3 and 4 only
(d) All four

Q3. Statement I: Galathea Bay in Great Nicobar Island is a primary nesting site for the Giant Leatherback Sea Turtle and the Nicobar Megapode. Statement II: The Galathea National Park and Megapode Wildlife Sanctuary have been denotified by the Government.

- **(a)** Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
- **(b)** Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
- **(c)** Statement I is correct but Statement II is incorrect
- **(d)** Statement I is incorrect but Statement II is correct

Q4. Which of the following correctly describes the three zones of a Biosphere Reserve under UNESCO's Man and the Biosphere Programme?

- **(a)** Core — strictly protected, no human activity; Buffer — research and education; Transition — sustainable use
- **(b)** Sanctuary — breeding grounds; Buffer — controlled tourism; Transition — agriculture
- **(c)** Reserve — notified by state; Buffer — notified by Centre; Transition — private land
- **(d)** Core — agroforestry permitted; Buffer — restricted; Transition — prohibited

Q5. Consider the following pairs of Biodiversity Hotspots and their Indian locations:

- **(a)** Western Ghats–Sri Lanka — Kerala and Tamil Nadu
- **(b)** Indo-Burma — Northeast India
- **(c)** Himalaya — Jammu & Kashmir, Uttarakhand, Sikkim, Arunachal Pradesh
- **(d)** Sundaland — Nicobar Islands

How many of the above pairs are correctly matched?

- **(a)** Only two
- **(b)** Only three
- **(c)** All four
- **(d)** Only one

ANSWERS WITH EXPLANATIONS

Q1. Answer: (b) 1, 3 and 4 only

Statement 2 is incorrect — Chilika was placed on the Montreux Record in 1993 but removed in 2002 after successful restoration by the Chilika Development Authority. Currently only Keoladeo and Loktak are on the Montreux Record.

Q2. Answer: (a) 1, 2 and 3 only

Statement 4 is incorrect — the first Tiger Reserve was Jim Corbett in Uttarakhand (1973), not Palamau. Corbett was among the original nine.

Q3. Answer: (b)

Both statements are individually correct. However, Statement II does NOT explain Statement I — the denotification came after the ecological importance was already known; it does not explain why the site is ecologically important.

Q4. Answer: (a)

Biosphere Reserves have three zones: Core (strict protection), Buffer (research/education), Transition (sustainable use). UNESCO's Man and the Biosphere Programme since 1971.

Q5. Answer: (c) All four

All four are correct. Four biodiversity hotspots partially in India: Western Ghats–Sri Lanka (west coast states); Indo-Burma (NE states, includes Andaman); Himalaya (J&K to Arunachal Pradesh); and Sundaland (Nicobar Islands only).

SECTION 05

Forests, Land & Soil

ISFR 2023, mangrove restoration, and the forest governance shift

Forests and land-use questions have risen steadily in UPSC Prelims, especially since 2022. Forest cover numbers, mangrove trends, and specific schemes (MISHTI, Green India Mission, Nagar Van) are repeatedly tested. This section covers the five most important developments in India's forest and land governance from the last 18 months.

MUST-KNOW

5.1 India State of Forest Report 2023 — the 18th edition

Prelims & Mains · Environment + Geography · Also: State PCS, CDS

The India State of Forest Report (ISFR) 2023 — the 18th biennial report by the Forest Survey of India (FSI) — was released by Environment Minister Bhupender Yadav at Dehradun in December 2024. It assesses forest and tree cover using remote sensing satellite data plus field-based National Forest Inventory (NFI). Total forest and tree cover now stands at 8,27,357 sq km — 25.17% of India's geographical area — up from 24.62% in 2021.

The report flags mixed signals. Forest cover rose marginally (+156 sq km) while tree cover grew sharply (+1,289 sq km). Forest carbon stock reached 7,285.5 million tonnes of carbon, which converts to 30.43 billion tonnes of CO_2 equivalent — reflecting an addition of 2.29 billion tonnes of CO_2 equivalent sink since the 2005 NDC baseline (target: 2.5–3 billion tonnes by 2030). But mangrove cover fell by 7.43 sq km, northeastern states continued decline, and the Western Ghats Ecologically Sensitive Area lost 58.22 sq km since 2013 (first-ever such assessment).

KEY FACTS FOR PRELIMS

Agency	Forest Survey of India (FSI), under MoEFCC; based in Dehradun; since 1981
Report series	Biennial since 1987; ISFR 2023 is the 18th edition
Forest & tree cover	8,27,357 sq km (25.17% of geographical area)
Forest cover alone	7,15,343 sq km (21.76%)
Tree cover alone	1,12,014 sq km (3.41%)
Change vs 2021	+156 sq km forest cover; +1,289 sq km tree cover; +1,445 sq km total
Top states (largest F&T cover)	Madhya Pradesh (85,724 sq km) > Arunachal Pradesh (67,083) > Maharashtra (65,383)
Top states (% forest cover)	Lakshadweep (91.33%) > Mizoram (85.34%) > Andaman & Nicobar (81.62%)
Mangrove cover	4,992 sq km (–7.43 sq km vs 2021); 12 states/UTs; Gujarat decreased
Carbon stock	7,285.5 Mt; 30.43 Bt CO2e total; +2.29 Bt sink since 2005 (target 2.5–3 Bt by 2030)
Forest fires	Top 3 fire-affected states: Uttarakhand > Odisha > Chhattisgarh

National Forest Policy 1988	Targets 33% of area under forest/tree cover; 19 states/UTs currently above 33%

REVISE: *Shankar IAS / PMF IAS — Forests chapter. Cross-reference with National Forest Policy 1988 and Forest Rights Act 2006.*
PYQ LINK: *UPSC 2018 and 2020 asked about ISFR findings and top states. Memorize top-3 lists for both absolute area and percentage.*

SHOULD-KNOW

5.2 Van (Sanrakshan Evam Samvardhan) Adhiniyam — the 2023 amendment in effect

Prelims & Mains · Environment + Polity · Also: State PCS

The Forest (Conservation) Amendment Act, 2023 — officially renamed Van (Sanrakshan Evam Samvardhan) Adhiniyam — came into force on 1 December 2023. It amends the parent Forest (Conservation) Act, 1980. Key changes: (i) defines categories of land to which the Act applies (notified forests and land recorded as forest in government records after 25 October 1980); (ii) exempts certain lands from the Act — land within 100 km of international borders, LoC, LAC for security/strategic projects; up to 10 hectares for security or defence projects; land for small public utility projects up to 5 hectares.

The amendment has been challenged in the Supreme Court. In February 2024, the SC directed that the 'deemed forest' definition from the 1996 T.N. Godavarman judgment continues to apply until states complete their own classifications. In October 2025, compensatory afforestation under the Van Adhiniyam became integrated with the Green Credit Programme, with MoEFCC notifying guidelines for using GCP afforestation to meet CA obligations.

KEY FACTS FOR PRELIMS

Act	Forest (Conservation) Amendment Act, 2023 = Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980
In force from	1 December 2023
Cut-off date	25 October 1980 for 'recorded forest' status
Key exemptions	Lands within 100 km of international border/LoC/LAC; up to 10 ha for security projects; up to 5 ha for utility projects in affected areas
SC order (Feb 2024)	Godavarman 'deemed forest' definition continues until states classify
October 2025 integration	Green Credit Programme afforestation may be used for Compensatory Afforestation
CAMPA	Compensatory Afforestation Fund Management and Planning Authority — National Authority 2016
T.N. Godavarman (1996)	SC expanded 'forest' to include any land satisfying dictionary definition of forest

REVISE: *Shankar IAS / PMF IAS — Environmental Laws. Laxmikanth for Supreme Court's environmental jurisprudence.*
PYQ LINK: *UPSC 2019 Mains asked about Forest Rights. Prelims questions on the 1980 Act's scope are likely given the new amendment.*

SHOULD-KNOW

5.3 MISHTI — Mangrove Initiative for Shoreline Habitats

Prelims · Environment + Economy · Also: State PCS

Mangrove Initiative for Shoreline Habitats & Tangible Incomes (MISHTI) was announced in Budget 2023–24 as a five-year programme (2023–2028) to restore and promote mangroves along India's coastline. It is implemented by MoEFCC with convergence from MGNREGS, CAMPA Fund, and other sources. Target: restore mangroves across 540 sq km spanning 11 states and 2 UTs.

MISHTI complements the MISHTI-partnered Mangrove Alliance for Climate (MAC) — a global initiative launched by UAE and Indonesia at COP27 with India as a founding member. India's mangrove cover (ISFR 2023) is 4,992 sq km; Gujarat has the largest area, followed by West Bengal (Sundarbans), Andaman & Nicobar, Andhra Pradesh. Mangroves are recognized as highly effective nature-based solutions for carbon sequestration and coastal protection.

KEY FACTS FOR PRELIMS

MISHTI	Announced Budget 2023–24; duration 2023–2028; MoEFCC-led
Target	540 sq km mangrove restoration across 11 states + 2 UTs
Convergence sources	MGNREGS + CAMPA + Coastal Zone Management + state funds
Global parallel	Mangrove Alliance for Climate (MAC) — launched COP27 by UAE & Indonesia; India founding member
Mangrove cover (ISFR 2023)	4,992 sq km; –7.43 sq km vs 2021
Top states for mangrove cover	Gujarat > West Bengal > Andaman & Nicobar > AP
Sundarbans	Largest single contiguous mangrove area in the world; UNESCO WHS + Biosphere Reserve + Ramsar site
CCTS offset methodology (2025)	Mangrove afforestation/reforestation is one of eight approved methodologies

REVISE: *Shankar IAS / PMF IAS — Coastal Ecosystems + Mangroves. NCERT Class XI Geography for coastal landforms.*
PYQ LINK: *UPSC 2018 asked about mangroves. UPSC 2022 asked about nature-based solutions. Mangroves are testable via both angles.*

SHOULD-KNOW

5.4 Soil health, desertification and the Rio Trio Initiative

Prelims · Environment + Geography · Also: State PCS

India's land degradation is serious: around 30% of India's geographical area is degraded, and ~70% of arable land is affected to some degree. Under UNCCD, India has pledged to achieve Land Degradation Neutrality (LDN) by 2030 and to restore 26 million hectares of degraded land under the Bonn Challenge.

At UNCCD COP16 (Riyadh 2024), the Rio Trio Initiative was launched to align the three Rio Conventions — UNCCD, UNFCCC, CBD — on nature-based solutions. India's major soil-health schemes include the Soil Health Card Scheme (2015), Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), and the National Mission on Natural Farming (NMNF, launched November 2024 with outlay Rs 2,481 crore). ISFR 2023 recorded improvements in Indian soil health: 87.16% shallow-to-deep soils (vs 83.53% in 2013), and soil organic carbon rose to 56.08 tonnes/hectare.

KEY FACTS FOR PRELIMS

India's degraded land	~30% of geographical area; ~70% of arable land affected
India's UNCCD commitments	Land Degradation Neutrality by 2030; 26 million hectares restored (Bonn Challenge)

Rio Trio Initiative	Launched at UNCCD COP16 Riyadh 2024; aligns UNCCD + UNFCCC + CBD
Bonn Challenge	Global pledge to restore 350 million ha of degraded land by 2030 (launched 2011)
Soil Health Card	Launched 2015; sampling and analysis every 3 years
National Mission on Natural Farming	Cabinet approval November 2024; Rs 2,481 crore outlay; promotes chemical-free, indigenous farming
ISFR 2023 soil findings	87.16% shallow-to-deep soils (+3.63%); SOC 56.08 t/ha (up from 55.85)
Desertification & Land Degradation Atlas	ISRO/SAC Space Applications Centre; latest edition 2021 (covers 2018–19 vs 2003–05)

REVISE: *Shankar IAS / PMF IAS — Soil + Desertification chapters. NCERT Class XI Geography for soil types.*
PYQ LINK: *UPSC 2023 asked about soil organic carbon. Land degradation figures and LDN are testable.*

GOOD-TO-KNOW

5.5 Green India Mission and forest-related schemes in news

Prelims · Environment · Also: State PCS

The National Mission for a Green India (GIM) is one of eight missions under the National Action Plan on Climate Change (NAPCC, 2008). Target: enhance forest/tree cover on 5 million hectares and improve quality on another 5 million ha. As of 2024–25, the Ministry had released Rs 944.48 crore to 17 states and 1 UT. Implemented through Joint Forest Management Committees (JFMCs).

Nagar Van Yojana (2020) focuses on urban forests in 1,000+ cities. School Nursery Yojana promotes tree planting in schools. Amrit Dharohar (2023) focuses on wetland conservation with community participation. Ek Ped Maa Ke Naam ('One Tree in Mother's Name') was launched by Prime Minister on World Environment Day (5 June 2024); by late 2024 over 80 crore saplings were planted.

KEY FACTS FOR PRELIMS

NAPCC (2008)	8 missions: GIM, Solar, Energy Efficiency, Sustainable Habitat, Water, Sustaining Himalayan Ecosystem, Sustainable Agriculture, Strategic Knowledge
GIM target	Enhance 5 million ha forest/tree cover; improve quality on another 5 million ha
Nagar Van Yojana 2020	546 projects approved across 31 states/UTs; Rs 431.77 crore allocated
Amrit Dharohar	Wetlands scheme; three-year programme 2023–26
Ek Ped Maa Ke Naam	Launched 5 June 2024 (World Environment Day); 140+ crore target
LiFE linkage	All campaigns align with Mission LiFE themes
Joint Forest Management (JFM)	Community-led forest management; JFMCs (Joint Forest Management Committees)

REVISE: *Shankar IAS / PMF IAS — Forest Management + NAPCC. Keep a one-page list of all 8 NAPCC missions.*
PYQ LINK: *UPSC 2020 asked about NAPCC missions. UPSC 2023 asked about Amrit Dharohar indirectly. Scheme-specific questions are standard.*

SECTION 5 — PRACTICE

Test yourself

Q1. With reference to the India State of Forest Report 2023, consider the following statements:

(a) It is the 18th such report, released biennially by the Forest Survey of India.
(b) Total forest and tree cover of the country is 25.17% of the geographical area.
(c) Mangrove cover increased by 296 sq km compared to the 2021 assessment.
(d) Madhya Pradesh has the largest forest and tree cover in the country.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1, 3 and 4 only
(c) 2 and 3 only
(d) All four

Q2. Match the following states/UTs with their forest cover percentage (ISFR 2023) in descending order:

(a) Lakshadweep — 91.33%
(b) Mizoram — 85.34%
(c) Andaman & Nicobar — 81.62%
(d) Arunachal Pradesh — 79.33%

How many of the above are correctly matched?

(a) Only two
(b) Only three
(c) All four
(d) Only one

Q3. With reference to the Mangrove Initiative for Shoreline Habitats & Tangible Incomes (MISHTI), consider:

(a) It was announced in Union Budget 2023–24.
(b) It aims to restore mangroves across 540 sq km in 11 states and 2 Union Territories.
(c) It draws convergence from MGNREGS and CAMPA Fund.
(d) India is a founding member of the Mangrove Alliance for Climate launched at COP27.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1, 3 and 4 only
(c) 2 and 4 only
(d) All four

Q4. Statement I: Under the Forest (Conservation) Amendment Act, 2023, lands within 100 km of an international border may be exempted from its application for strategic projects. Statement II: The Amendment Act exempts such lands because the parent Act of 1980 explicitly permits strategic use of border areas.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Q5. Which of the following missions are part of the National Action Plan on Climate Change (2008)?

(a) National Solar Mission
(b) National Mission for Enhanced Energy Efficiency
(c) National Mission for Sustaining the Himalayan Ecosystem
(d) National Mission on Quantum Technologies

Select the correct answer using the code given below:

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) 2, 3 and 4 only
(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (a) 1, 2 and 4 only

Statement 3 is incorrect — mangrove cover actually DECREASED by 7.43 sq km compared to 2021 (to 4,992 sq km); Gujarat saw the sharpest decline. The 296 sq km figure refers to an earlier report cycle. All other statements are correct.

Q2. Answer: (b) Only three

The first three pairs (Lakshadweep, Mizoram, Andaman & Nicobar) are correct per ISFR 2023. Arunachal Pradesh has ~79% forest cover per ISFR 2023 — the figure given is plausible but not exactly matched in the official data. Only three pairs are confirmed correctly.

Q3. Answer: (d) All four

All four statements are correct. MISHTI was announced in Budget 2023–24 (February 2023), targets 540 sq km across 11 states + 2 UTs, draws from MGNREGS/CAMPA, and India is a founding member of the Mangrove Alliance for Climate (launched COP27 Egypt by UAE and Indonesia).

Q4. Answer: (c)

Statement I is correct — the 2023 Amendment exempts lands up to 100 km from international borders/LoC/LAC for strategic/security projects. Statement II is incorrect — the parent 1980 Act did NOT explicitly permit strategic border use; this exemption is a NEW introduction by the 2023 Amendment.

Q5. Answer: (b) 1, 2 and 3 only

NAPCC (2008) has 8 missions: Solar, Enhanced Energy Efficiency, Sustainable Habitat, Water, Sustaining Himalayan Ecosystem, Green India, Sustainable Agriculture, Strategic Knowledge on Climate. Quantum Technologies is a separate National Mission (2023), not under NAPCC.

SECTION 06

Pollution & Waste Management

CAQM, GRAP, plastic rules, and EPR compliance

Pollution-related questions in UPSC Prelims focus less on science and more on the governance framework — which authority, which rule, which schedule. This section covers the five most important developments from October 2024 to March 2026 in India's pollution and waste management regime.

MUST-KNOW

6.1 CAQM, GRAP Stage IV, and Delhi's air pollution crisis

Prelims · Environment + Polity · Also: State PCS

The Commission for Air Quality Management (CAQM) in the National Capital Region and Adjoining Areas was established by Parliament in 2021 as a statutory body. It replaced the Supreme Court-mandated EPCA and subsumed over a dozen earlier bodies. CAQM has powers of direction — its orders prevail over any state law — and can impose penalties. In 2023 the NCR was expanded to include parts of Uttar Pradesh, Haryana, and Rajasthan.

In November 2024, CAQM invoked Graded Response Action Plan (GRAP) Stage IV across NCR as the AQI crossed 450 ('Severe+'). Stage IV measures include banning of non-essential trucks, suspension of construction, closure of schools for primary classes, and work-from-home advisory. In early 2025, CAQM tightened norms on the BS-VI implementation and end-of-life vehicles. Stubble burning in Punjab and Haryana remains the most visible seasonal contributor alongside vehicular and industrial sources.

KEY FACTS FOR PRELIMS

CAQM	Commission for Air Quality Management Act 2021; statutory body; subsumes earlier bodies
Jurisdiction	NCR + adjoining areas in Punjab, UP, Haryana, Rajasthan
GRAP stages	I (Poor, AQI 201–300); II (Very Poor, 301–400); III (Severe, 401–450); IV (Severe+, >450)
GRAP IV measures	Non-essential trucks banned; construction suspended; schools closed for younger classes; WFH advisory
AQI bands	Good 0–50; Satisfactory 51–100; Moderate 101–200; Poor 201–300; Very Poor 301–400; Severe 401–500
Pollutants in AQI (8)	PM2.5, PM10, NO2, SO2, CO, O3, NH3, Pb
Stubble burning	Punjab + Haryana; seasonal October–November; CPCB Commission monitors
Delhi 2024 stubble-burning season	AQI peaked at 494 in November; GRAP IV imposed
IQAir World AQ Report 2024	Delhi ranked most polluted capital city (8th consecutive year); India 3rd most polluted country (after Bangladesh and Pakistan)

REVISE: *Shankar IAS / PMF IAS — Air Pollution chapter. Laxmikanth for CAQM as a statutory body. AQI calculation method is testable.*

PYQ LINK: *UPSC 2016 asked about National AQI. UPSC 2020 asked about CPCB. GRAP stages and triggers are classic Statement-Based question material.*

MUST-KNOW

6.2 Plastic Waste Management Rules 2024 amendments + EPR

Prelims · Environment + Governance · Also: State PCS

The Plastic Waste Management (Amendment) Rules, 2024 (notified January 2024) further strengthened India's plastic waste regime. Since 1 July 2022, 19 single-use plastic items were banned (thickness < 75 microns for carry bags since 2022, further raised to 120 microns from 31 December 2022). The 2024 amendments expanded the definition of 'biodegradable plastic,' tightened labelling rules, and brought more products under Extended Producer Responsibility (EPR).

EPR targets apply to Producers, Importers, and Brand Owners (PIBOs) for four categories: rigid, flexible, multilayered, and compostable plastic packaging. The Central Pollution Control Board (CPCB) manages an online centralized EPR portal where PIBOs register and buy EPR certificates from registered recyclers. In 2024–25, new sector coverage expanded to include agarbatti, gutka, and cigarette packaging. Penalty structure under the Environment Protection Act was reinforced.

KEY FACTS FOR PRELIMS

Parent Rules	Plastic Waste Management Rules, 2016 (under Environment Protection Act, 1986)
Major amendments	2021 + 2022 (SUP ban) + 2024 (biodegradable plastic + EPR strengthening)
Carry bag thickness	Minimum 120 microns from 31 Dec 2022 (earlier 75)
19 banned single-use items	Includes plastic straws, cutlery, earbuds with plastic stems, plastic sticks for balloons, flags, PVC banners <100 microns, etc.
EPR targets (4 categories)	Category I (rigid), II (flexible), III (multilayered), IV (compostable)
EPR implementing body	Central Pollution Control Board (CPCB) via centralized online portal
PIBO	Producers + Importers + Brand Owners — all covered under EPR
2024 additions	Expanded to cover agarbatti, gutka, cigarette packaging
India's plastic waste	~3.4 million tonnes per year generated (CPCB); recycling rate ~60% (disputed estimate)

REVISE: *Shankar IAS / PMF IAS — Pollution chapter + Environment Protection Act 1986 chapter. Memorize list of 19 banned SUP items.*

PYQ LINK: *UPSC 2019 asked about plastic recycling. UPSC 2023 asked about single-use plastics. EPR as a concept is testable directly.*

SHOULD-KNOW

6.3 E-Waste Management Rules 2022 and EPR

Prelims · Environment · Also: State PCS

The E-Waste (Management) Rules, 2022 came into force on 1 April 2023, replacing the 2016 Rules. The scope has been significantly expanded — now covering 106 electrical and electronic equipment (EEE) types (earlier 21). Extended Producer Responsibility (EPR) obligations apply to manufacturers, producers, refurbishers, and recyclers, managed via an online portal operated by CPCB.

New features: producers must have EPR registration; annual recycling targets escalate from 60% (2023–24) to 80% (2026–27); registration is a prerequisite to sell EEE in India; non-compliance attracts environmental compensation. Mercury-added products, lithium-ion batteries, and solar panels are brought under specific provisions. India's e-waste generation is ~3.8 million tonnes (2024), the third-largest globally.

KEY FACTS FOR PRELIMS

E-Waste Rules 2022	In force from 1 April 2023
Scope	106 EEE types (earlier 21 under 2016 Rules)
Regulator	Central Pollution Control Board (CPCB) — online EPR portal
EPR targets	60% (2023–24) → 65% (2024–25) → 70% (2025–26) → 80% (2026–27)
India's e-waste	~3.8 million tonnes/year (2024); 3rd globally after China, US
New coverage	Solar PV panels; lithium-ion batteries; mercury-added products
Related: Battery Waste Rules 2022	Separate EPR regime; covers all battery types incl. EV batteries
Related: Hazardous Waste Rules	Implements Basel Convention; updated in 2022 and 2023

REVISE: *Shankar IAS / PMF IAS — Waste Management chapter. Note the different EPR regimes (Plastic, E-Waste, Battery, Tyre).*
PYQ LINK: *UPSC 2018 asked about e-waste. UPSC 2022 asked about battery waste. Recent rules updates are testable.*

SHOULD-KNOW

6.4 Water pollution, Ganga basin and Jal Jeevan Mission

Prelims · Environment + Governance · Also: State PCS, Banking

Namami Gange Programme (2014) — the flagship for rejuvenating the Ganga — has been extended to 2026 with an approved outlay of Rs 22,500 crore (Second Phase). The National Mission for Clean Ganga (NMCG) is the implementing body. In 2024–25, Mission focus shifted to sewage treatment capacity (targeting 5,500+ MLD additional capacity), afforestation along the basin, and livelihood integration.

Jal Jeevan Mission (2019) — tap water connection to every rural household — had connected 15 crore rural households by March 2025 (vs 3.23 crore at launch). The original 2024 deadline has been extended to 2028. Jal Shakti Abhiyan (2019, renamed 'Catch the Rain' campaign in 2021) continues annually. The Central Ground Water Authority (CGWA) oversees groundwater regulation; 14% of assessed blocks are 'over-exploited' as per the 2023 assessment.

KEY FACTS FOR PRELIMS

Namami Gange	2014; second phase 2021–2026; Rs 22,500 crore; NMCG (constituted as authority in 2016)
Jal Jeevan Mission	Launched 15 Aug 2019; extended to 2028; 15 crore connections by March 2025
Funding share	50:50 Centre-State (90:10 for NE, Himalayan states, J&K, Ladakh)
Jal Shakti Abhiyan	Launched 2019; 'Catch the Rain' since 2021; annual monsoon-season campaign
Ground water assessment (2023)	14% of blocks 'over-exploited'; 4% 'critical'
Atal Bhujal Yojana	World Bank-assisted; groundwater management in 7 states

Water (Prevention & Control of Pollution) Act	1974 — India's first dedicated pollution control legislation; Wildlife Protection Act 1972 preceded it as the first wildlife conservation law

REVISE: *Shankar IAS / PMF IAS — Water Pollution + Water Resources. Remember: Water Act 1974 preceded EPA 1986.*
PYQ LINK: *UPSC 2022 asked about Namami Gange. Jal Jeevan Mission progress numbers are testable.*

GOOD-TO-KNOW

6.5 Solid waste, CPCB, and the 2024 IPCC methane/waste update

Prelims · Environment · Also: State PCS

Solid Waste Management Rules, 2016 (amended 2024) mandate source segregation into wet, dry, and hazardous. Implementation is weakest in cities with populations under 1 lakh. Swachh Bharat Mission–Urban 2.0 (SBM-U 2.0, launched 2021) targets 'garbage-free' cities with 100% source segregation, 100% door-to-door collection, and 100% scientific processing by 2026.

The Central Pollution Control Board (CPCB) is a statutory body under the Water Act 1974 and EPA 1986. It coordinates with State Pollution Control Boards. In 2024, CPCB approved new standards for Construction & Demolition (C&D) waste and updated norms for bioremediation of legacy waste dumps. India's 3,500+ legacy waste dumpsites are a major national concern; the Dhapa (Kolkata), Deonar (Mumbai), and Ghazipur (Delhi) dumps feature regularly in news.

KEY FACTS FOR PRELIMS

Solid Waste Management Rules 2016	Amended 2024; source segregation mandatory
SBM-U 2.0	Launched 1 Oct 2021; outlay Rs 1.41 lakh crore; timeline to 2026
CPCB	Statutory body under Water Act 1974 + EPA 1986; HQ Delhi; under MoEFCC
State Pollution Control Boards (SPCBs)	Established under Water Act 1974
C&D Waste Rules 2016	Require separate handling of construction & demolition waste
Legacy dumpsites	~3,500+ across India; focus of SBM-U 2.0 bioremediation
Iconic dumps in news	Dhapa (Kolkata), Deonar (Mumbai), Ghazipur (Delhi)
India's MSW generation	~1.70 lakh tonnes/day (2024); ~77% treated according to MoHUA

REVISE: *Shankar IAS / PMF IAS — Waste Management + Pollution. Laxmikanth for statutory bodies.*
PYQ LINK: *UPSC 2020 asked about CPCB. UPSC 2018 asked about waste management. Scheme names + timelines are standard Prelims material.*

SECTION 6 — PRACTICE

Test yourself

Q1. With reference to the Commission for Air Quality Management (CAQM) in NCR and Adjoining Areas, consider:

(a) It was constituted by an Act of Parliament in 2021.
(b) Its jurisdiction includes parts of Punjab, Haryana, Rajasthan, and Uttar Pradesh.
(c) Its orders prevail over state government orders in case of conflict.

(d) It replaced the Environment Pollution (Prevention & Control) Authority (EPCA).

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1, 2, 3 and 4
(d) 1, 3 and 4 only

Q2. Consider the following statements regarding the Graded Response Action Plan (GRAP) for Delhi-NCR:

(a) GRAP is implemented by the CAQM.
(b) Stage IV of GRAP is triggered when the AQI crosses 450 (Severe+).
(c) Under Stage III, primary schools are closed and construction activities are suspended.
(d) GRAP has four stages corresponding to Poor, Very Poor, Severe, and Severe+ AQI categories.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1, 3 and 4 only
(c) 2 and 3 only
(d) All four

Q3. Which of the following pollutants are included in the National Air Quality Index (NAQI)?

(a) PM2.5 and PM10
(b) NO2 and SO2
(c) Ozone and Carbon Monoxide
(d) Ammonia and Lead

Select the correct answer using the code given below:

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) 1, 2, 3 and 4
(d) None of these

Q4. Statement I: The Central Pollution Control Board (CPCB) is a statutory body under the Environment (Protection) Act, 1986. Statement II: The CPCB was first established under the Water (Prevention and Control of Pollution) Act, 1974.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Q5. With reference to Extended Producer Responsibility (EPR) in India, consider:

(a) EPR applies to Producers, Importers, and Brand Owners (PIBOs) of plastic packaging.
(b) The E-Waste (Management) Rules, 2022 cover 106 types of electrical and electronic equipment.
(c) The EPR targets under the Plastic Waste Management Rules are administered by the Ministry of Consumer Affairs.
(d) EPR targets under E-Waste Rules rise progressively from 60% in 2023–24 to 80% by 2026–27.

Which of the statements given above are correct?

(a) 1, 2 and 4 only
(b) 1, 3 and 4 only
(c) 2, 3 and 4 only
(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (c) All four
All four are correct. CAQM was constituted by an Act of Parliament in 2021 (Commission for Air Quality Management Act 2021). It has jurisdiction over NCR + adjoining areas (parts of Punjab, Haryana, Rajasthan, UP). Its orders override state orders. It replaced the Supreme Court-mandated EPCA (Environment Pollution Prevention and Control Authority).

Q2. Answer: (a) 1, 2 and 4 only
Statement 3 is incorrect — primary school closure and construction suspension fall under GRAP Stage IV (Severe+, AQI >450), not Stage III. Stage III covers AQI 401–450 with stricter measures like ban on private diesel vehicles but schools and construction are partially affected.

Q3. Answer: (c) 1, 2, 3 and 4
All eight pollutants are part of the National AQI: PM2.5, PM10, NO2, SO2, CO, O3 (Ozone), NH3 (Ammonia), and Pb (Lead). The AQI is based on 24-hour averages (except O3 and CO which use 8-hour averages).

Q4. Answer: (d)
Statement I is incorrect — CPCB was established under the Water (Prevention and Control of Pollution) Act, 1974, not under EPA 1986. Statement II is correct. The Air Act 1981 and EPA 1986 later gave CPCB additional responsibilities, but its original statutory basis is the 1974 Water Act.

Q5. Answer: (a) 1, 2 and 4 only
Statement 3 is incorrect — EPR under Plastic Waste Management Rules is administered by MoEFCC (via CPCB), not the Ministry of Consumer Affairs. All other statements are correct.

SECTION 07

Renewable Energy & Clean Tech

PM Surya Ghar, nuclear push, offshore wind, and the 500 GW target

UPSC increasingly tests energy transition — not just as environment, but as economy, governance, and international relations. This section covers the five most important developments in India's clean energy push from October 2024 to March 2026.

MUST-KNOW

7.1 PM Surya Ghar: Muft Bijli Yojana — rooftop solar for 1 crore homes

Prelims · Economy + Environment · Also: State PCS, SSC

PM Surya Ghar: Muft Bijli Yojana (Pradhan Mantri Surya Ghar Muft Bijli Yojana) was launched by Prime Minister Modi on 13 February 2024 (after an interim budget announcement on 1 February 2024). The scheme aims to install rooftop solar on 1 crore households by FY2026–27 and provide up to 300 units of free electricity every month. Total outlay is Rs 75,021 crore.

Central Financial Assistance (CFA): Rs 30,000 per kW for up to 2 kW; Rs 18,000/kW for next 1 kW (capped at Rs 78,000 for 3 kW+ systems). The scheme includes a Model Solar Village in every district and a 'Renewable Energy Service Company' (RESCO) option. Collateral-free loans up to Rs 2 lakh from scheduled commercial banks at ~7% interest. By March 2026, over 14 lakh households had installed rooftop solar under the scheme.

KEY FACTS FOR PRELIMS

Launched	13 February 2024 by PM Modi
Target	1 crore households with rooftop solar by FY2026–27
Outlay	Rs 75,021 crore
Benefit	Up to 300 units/month free electricity
CFA for up to 2 kW	Rs 30,000 per kW
CFA maximum (≥3 kW)	Rs 78,000 (capped)
Loan facility	Collateral-free, up to Rs 2 lakh, ~7% interest
Nodal Ministry	Ministry of New and Renewable Energy (MNRE)
Implementing agency	National Portal administered by MNRE; DISCOMs process applications

REVISE: *Shankar IAS / PMF IAS — Solar Energy chapter. Cross-reference with International Solar Alliance and India's 500 GW target.*

PYQ LINK: *UPSC 2018 asked about solar rooftop. New schemes like PM Surya Ghar are almost guaranteed Prelims material.*

MUST-KNOW

7.2 500 GW non-fossil capacity by 2030 — installed capacity progress

Prelims & Mains · Economy + Environment · Also: CDS

India's flagship climate target is 500 GW of non-fossil-fuel-based electricity generation capacity by 2030 — announced by PM Modi at COP26 Glasgow (Panchamrit). As of early 2026, India's installed non-fossil capacity has crossed 230 GW — comprising large hydro, solar, wind, biomass, small hydro, and nuclear. Total installed capacity is ~460 GW.

India has already achieved (a year ahead) the 2015 NDC target of 40% cumulative non-fossil installed capacity. Renewable energy alone (solar + wind + biomass + small hydro) crossed 200 GW in 2025. Solar leads the mix with ~90+ GW installed (up from 2.6 GW in 2014). India ranks 4th globally in installed renewable energy capacity (after China, US, Germany — per IRENA 2024). Wind power is ~47 GW; India ranks 4th globally for wind.

KEY FACTS FOR PRELIMS

Panchamrit target	500 GW non-fossil capacity by 2030
Updated NDC target	~50% cumulative installed electric power from non-fossil sources by 2030
Current status (early 2026)	Non-fossil capacity ~230 GW; total capacity ~460 GW
Solar installed	~90+ GW; up from 2.6 GW in 2014
Wind installed	~47 GW
Global rank (renewables)	4th (after China, US, Germany) per IRENA 2024
Solar parks scheme	50+ solar parks; Bhadla (Rajasthan, ~2,245 MW), Pavagada (Karnataka, 2,050 MW) are largest
2070 net-zero target	Requires ~7,000 GW renewable capacity (CEEW/NITI Aayog estimate)
Per capita emissions	~2 t CO2/year — one-third of global average (~6 t); one-seventh of US

REVISE: *Shankar IAS / PMF IAS — Renewable Energy + Panchamrit. Keep a sector-wise installed capacity table (Solar, Wind, Hydro, Nuclear, Biomass).*

PYQ LINK: *UPSC 2017 Mains asked about India's renewable energy progress. UPSC 2023 asked about solar capacity. Classic recurring theme.*

SHOULD-KNOW

7.3 International Solar Alliance and solar diplomacy

Prelims · Environment + International Affairs · Also: State PCS, CDS

The International Solar Alliance (ISA) — co-founded by India and France at COP21 Paris in 2015 — became a treaty-based intergovernmental organization in 2017. It has headquarters in Gurugram, Haryana; the ISA Secretariat is located at the National Institute of Solar Energy (NISE), Gwal Pahari. As of 2025, ISA has 122 signatory and member countries (after amending its Framework Agreement in 2020 to open membership to all UN member states).

Major 2024–25 developments: the 7th ISA Assembly (Delhi, November 2024) approved scaling-up of 'Solar for Schools' and 'Solar for Health' programmes. India and ISA launched the 'Solar for SIDS' initiative for Small Island Developing States. ISA is the only intergovernmental organization outside the UN system headquartered in India — a significant soft-power achievement.

KEY FACTS FOR PRELIMS

ISA	International Solar Alliance — treaty organization since 2017

Co-founders	India and France; announced at COP21 Paris, November 2015
Headquarters	Gurugram, Haryana — only intergovernmental body HQ outside UN system in India
Members	122 signatory countries (opened to all UN members after 2020 amendment)
7th Assembly	Delhi, 3–6 November 2024
Key initiatives	Solar for Schools + Solar for Health + Solar for SIDS + One Sun One World One Grid (OSOWOG)
Funding pledge target	Mobilize USD 1 trillion for solar by 2030

REVISE: *Shankar IAS / PMF IAS — Solar Energy + International organizations. Note ISA as an Indian foreign policy instrument.*
PYQ LINK: *UPSC 2016 asked about ISA. UPSC 2022 Mains asked about Indian solar diplomacy. Always testable.*

SHOULD-KNOW

7.4 Nuclear power, Bharat Small Reactor (BSR), and Amrit Kaal vision

Prelims · Economy + Environment + Science & Tech · Also: CDS

India's installed nuclear capacity is ~8.18 GW across 24 operational reactors (as of 2025). Budget 2024–25 announced a 'Nuclear Energy Mission' targeting 100 GW of nuclear capacity by 2047 as part of Amrit Kaal vision. Budget 2025–26 launched the Bharat Small Reactor (BSR) initiative — indigenous small modular reactors (SMRs) of ~200 MWe, with public-private partnership. Rs 20,000 crore corpus was earmarked for nuclear research.

India's nuclear programme follows the 'three-stage' plan proposed by Dr. Homi Bhabha: (1) Pressurized Heavy Water Reactors (PHWRs) using natural uranium; (2) Fast Breeder Reactors using plutonium + thorium; (3) Advanced Heavy Water Reactors using U-233 from thorium. The Prototype Fast Breeder Reactor (PFBR) at Kalpakkam, Tamil Nadu — critical milestone for Stage 2 — achieved first criticality in March 2024.

KEY FACTS FOR PRELIMS

Current installed nuclear capacity	~8.18 GW (24 reactors operational)
2030 target	~22 GW per earlier plan
Amrit Kaal target	100 GW of nuclear capacity by 2047 (Budget 2024–25)
Bharat Small Reactor	~200 MWe indigenous SMR; Budget 2025–26 announced PPP; Rs 20,000 crore corpus
Three-stage programme	Stage 1: PHWR (natural U); Stage 2: FBR (Pu + Th); Stage 3: AHWR (U-233 from Th)
PFBR Kalpakkam	Prototype Fast Breeder Reactor; first criticality March 2024
Thorium reserves	India has ~25% of world's thorium reserves (Kerala beach sands)
Nodal body	Department of Atomic Energy (DAE); Nuclear Power Corporation of India Ltd (NPCIL)

REVISE: *Shankar IAS / PMF IAS — Energy chapter. NCERT Class XII Physics for nuclear reactions (fission vs fusion).*
PYQ LINK: *UPSC 2019 asked about FBR. UPSC 2023 asked about thorium. BSR is new and highly likely for 2026 Prelims.*

SHOULD-KNOW

7.5 Offshore wind, pumped storage, EV push, and E20 ethanol

Prelims · Economy + Environment · Also: State PCS

India announced its first Offshore Wind Energy Lease Rules in 2023. In October 2024, the first set of lease tenders for 4 GW offshore wind blocks off Tamil Nadu and Gujarat were notified. Viability Gap Funding of Rs 7,453 crore was approved by Cabinet for 1 GW of offshore capacity (June 2024). India's potential is estimated at 70 GW (36 GW off TN; 34 GW off Gujarat).

Pumped Storage Projects (PSPs) — large-scale energy storage using reservoir pumping — have become central to India's RE integration. Union Cabinet approved Rs 12,000 crore viability gap funding for PSPs in April 2025. Target: 50 GW of PSP by 2047. On EVs: FAME-II ended December 2023; PM E-Drive Scheme was launched October 2024 with Rs 10,900 crore outlay. E20 (20% ethanol-blended petrol) target was achieved in 2025 — five years ahead of the 2030 deadline.

KEY FACTS FOR PRELIMS

Offshore wind rules	2023 Lease Rules; first tenders October 2024 for 4 GW off TN + Gujarat
Offshore wind VGF	Rs 7,453 crore approved June 2024 for 1 GW
India's offshore wind potential	~70 GW (36 GW off TN coast + 34 GW off Gujarat coast)
Pumped Storage Projects (PSP)	Cabinet VGF Rs 12,000 cr April 2025; target 50 GW by 2047
PM E-Drive Scheme	Launched October 2024; Rs 10,900 crore; succeeds FAME-II
E20 ethanol	20% ethanol blended petrol; achieved 2025 (5 years ahead of 2030 target)
National Green Hydrogen Mission	Approved January 2023; 5 MMT/year target by 2030
Ethanol feedstock	Sugarcane, rice grain, maize; 1G and 2G (second-generation cellulosic) plants

REVISE: *Shankar IAS / PMF IAS — Renewable Energy + Transport emissions. Keep a one-page sheet of all major RE schemes and their ministries.*

PYQ LINK: *UPSC 2020 asked about ethanol blending. UPSC 2023 asked about EVs. New scheme names are recurring Prelims material.*

SECTION 7 — PRACTICE

Test yourself

Q1. With reference to PM Surya Ghar: Muft Bijli Yojana, consider:

- **(a)** It was launched by the Prime Minister in February 2024.
- **(b)** It aims to install rooftop solar on 1 crore households by 2026–27.
- **(c)** It provides up to 300 units of free electricity per month to beneficiary households.
- **(d)** Central Financial Assistance is capped at Rs 78,000 per household for 3 kW or larger systems.

Which of the statements given above are correct?

- **(a)** 1 and 3 only
- **(b)** 2, 3 and 4 only

(c) 1, 3 and 4 only
(d) All four

Q2. With reference to the International Solar Alliance (ISA), consider:

(a) It was co-founded by India and France at the COP21 Paris climate summit.
(b) It is the only intergovernmental organization outside the UN system headquartered in India.
(c) Membership is open to all UN member states after a 2020 amendment.
(d) Its headquarters are in New Delhi.

Which of the statements given above are correct?

(a) 1 and 4 only
(b) 1, 2 and 3 only
(c) 2 and 3 only
(d) All four

Q3. Consider the following statements regarding India's nuclear programme:

(a) The Prototype Fast Breeder Reactor (PFBR) at Kalpakkam is part of Stage 2 of India's three-stage nuclear programme.
(b) Stage 3 of the programme envisages use of U-233 obtained from thorium.
(c) The Bharat Small Reactor initiative was announced in Union Budget 2025–26.
(d) The Nuclear Power Corporation of India Ltd (NPCIL) is under the Ministry of New and Renewable Energy.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1 and 4 only
(c) 2, 3 and 4 only
(d) All four

Q4. Statement I: India achieved its E20 (20% ethanol blending in petrol) target in 2025, five years ahead of the 2030 deadline. Statement II: Ethanol blending in petrol reduces vehicular emissions of CO2 and certain particulate matter.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Q5. With reference to India's 500 GW non-fossil capacity target by 2030, consider:

(a) The target was announced by the Prime Minister at COP26 Glasgow as part of Panchamrit.
(b) As of early 2026, India's installed non-fossil capacity has crossed 230 GW.
(c) India's solar installed capacity is higher than its wind installed capacity.
(d) India ranks 1st globally in installed renewable energy capacity.

Which of the statements given above are correct?

(a) 1, 2 and 3 only
(b) 1, 2 and 4 only
(c) 2 and 4 only
(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (d) All four

All four are correct. PM Surya Ghar was launched on 13 February 2024; target is 1 crore households by FY2026–27; up to 300 units free; CFA is capped at Rs 78,000 for systems ≥3 kW (Rs 30,000/kW for first 2 kW + Rs 18,000 for third kW).

Q2. Answer: (b) 1, 2 and 3 only

Statement 4 is incorrect — ISA Secretariat is in Gurugram, Haryana (at the NISE campus in Gwal Pahari), not Delhi. All other statements are correct. ISA is indeed the first intergovernmental body outside the UN system headquartered in India.

Q3. Answer: (a) 1, 2 and 3 only

Statement 4 is incorrect — NPCIL is under the Department of Atomic Energy (DAE), which reports directly to the Prime Minister's Office, NOT under the Ministry of New and Renewable Energy. MNRE handles solar, wind, hydrogen, etc.

Q4. Answer: (b)

Both statements are correct. Statement I confirmed — India achieved E20 in 2025, 5 years ahead of the 2030 target. Statement II is also correct — ethanol blending reduces certain emissions (though with caveats about NOx and aldehydes). However, II is NOT the 'explanation' for I — the target was set for policy/climate reasons; the environmental benefits are the motivation, not the explanation for early achievement.

Q5. Answer: (a) 1, 2 and 3 only

Statement 4 is incorrect — India ranks 4th globally in installed renewable energy capacity (after China, US, Germany) per IRENA. Solar (~90+ GW) is indeed higher than wind (~47 GW).

SECTION 08

Reports, Indices & Governance

UNEP, WMO, WWF, IPCC — the global reporting architecture

Every year UPSC tests the source of a report — who publishes it, what it measures, where India ranks. This section consolidates the six most important international and Indian environmental reports and indices from the last 18 months. Learn the source organization, the theme, and India's rank or finding.

MUST-KNOW

8.1 UNEP Emissions Gap Report — warming trajectory updates

Prelims · Environment · Also: State PCS

The UN Environment Programme's Emissions Gap Report — released annually before each COP — measures the gap between current emission pledges and what would keep warming below 1.5°C or 2°C. The 2024 edition (titled 'No More Hot Air … Please!'), released before COP29 Baku, projected warming of 2.6–3.1°C above pre-industrial by 2100 if current policies continue. The 2025 edition before COP30 Belém projected a somewhat improved 2.3–2.5°C trajectory — reflecting improving policies but still far from 1.5°C.

Key finding: global GHG emissions reached a new record of 57.1 GtCO2e in 2023. G20 countries account for 77% of global emissions. For 1.5°C pathway, emissions must fall 42% by 2030 compared to 2019 levels; for 2°C pathway, 28%. The report frames the urgent need for stronger NDCs in the post-2030 cycle (next NDCs due 2025–26 before COP31).

KEY FACTS FOR PRELIMS

Publisher	United Nations Environment Programme (UNEP); HQ Nairobi, Kenya
Release schedule	Annually before each UNFCCC COP
2024 edition	Released October 2024 before COP29; warming trajectory 2.6–3.1°C
2025 edition	Released October 2025 before COP30; trajectory 2.3–2.5°C (improvement)
Global emissions 2023	57.1 GtCO2e — new record high
G20 share	77% of global emissions
1.5°C cut needed	42% below 2019 levels by 2030
2°C cut needed	28% below 2019 levels by 2030
Linked report	UNEP Adaptation Gap Report — shows adaptation finance at USD 26 bn (2023), far below USD 310 bn/yr needs

REVISE: *Shankar IAS / PMF IAS — Climate Change chapter. Memorize: who publishes what — UNEP vs UNFCCC vs WMO vs IPCC.*

PYQ LINK: *UPSC 2018 asked about Emissions Gap Report. UPSC 2022 asked about UNEP. Publisher-report pairings are classic Prelims material.*

SHOULD-KNOW

8.2 IPCC AR7 cycle and WMO State of the Global Climate Report

Prelims · Environment · Also: State PCS

The Intergovernmental Panel on Climate Change (IPCC) — a joint body of UNEP and WMO set up in 1988 — completed its Sixth Assessment Report (AR6) in 2023. The Seventh Assessment Report (AR7) cycle began in 2023; Working Group I contributions are expected 2027–28. In 2024, IPCC released a Methodology Report on Short-Lived Climate Forcers and a Special Report on Cities. IPCC's AR6 famously established that global warming reached ~1.1°C above pre-industrial by 2011–20; AR7 is likely to confirm 1.5°C crossed in a single-year basis.

The World Meteorological Organization (WMO) State of the Global Climate Report 2024 (released March 2025) confirmed 2024 as the warmest year on record, with global mean temperature approximately 1.55°C above pre-industrial for the first time on an annual basis. The report found ocean heat content at a record, sea ice extents near record lows, and glaciers losing mass for the 37th consecutive year.

KEY FACTS FOR PRELIMS

IPCC	Joint body of UNEP + WMO, set up 1988; HQ Geneva
Nobel Prize	IPCC + Al Gore received 2007 Nobel Peace Prize
AR6	Completed 2023; WG1 (Physical Science), WG2 (Impacts), WG3 (Mitigation), Synthesis Report
AR7 cycle	Started 2023; outputs expected 2027–28
WMO State of Climate 2024	Released March 2025; 2024 confirmed warmest year on record
1.55°C breach	2024 global mean temperature ~1.55°C above pre-industrial (annual basis, first time)
Glacier mass loss	37 consecutive years of glacier mass loss (through 2024)
Ocean heat content	Record high in 2024 per WMO
UNFCCC / IPCC / WMO distinction	UNFCCC = treaty body (secretariat Bonn); IPCC = scientific assessment; WMO = UN specialized agency

REVISE: *Shankar IAS / PMF IAS — IPCC + Climate Change chapters. Always know: publisher, founding year, and HQ for each.*
PYQ LINK: *UPSC 2015 asked about IPCC. UPSC 2020 asked about WMO. Publisher questions recur every 2 years.*

SHOULD-KNOW

8.3 WWF Living Planet Report 2024 — 73% wildlife decline

Prelims · Environment · Also: State PCS

WWF's Living Planet Report 2024 (released October 2024) found that monitored wildlife populations have declined by 73% on average between 1970 and 2020 — based on the Living Planet Index (LPI), which tracks nearly 35,000 population trends for 5,495 vertebrate species. Freshwater populations show the steepest decline (85%), followed by terrestrial (69%) and marine (56%). Latin America and the Caribbean show the worst regional decline (95%).

The report warns that cascading tipping points — Amazon rainforest dieback, polar ice loss, coral die-off, North Atlantic circulation collapse — are nearing irreversible thresholds. India is flagged as facing particular pressure on freshwater ecosystems and pollinator populations. WWF was founded in 1961; its biennial Living Planet Report started in 1998. The Zoological Society of London (ZSL) maintains the LPI database.

KEY FACTS FOR PRELIMS

Publisher	World Wide Fund for Nature (WWF) — founded 1961
Partner	Zoological Society of London (ZSL) — maintains Living Planet Index database
Frequency	Biennial; first edition 1998
2024 edition headline	73% decline in monitored wildlife populations (1970–2020)
Freshwater decline	85% (steepest of any system)
Terrestrial decline	69%
Marine decline	56%
Worst region	Latin America + Caribbean (95% decline)
Five tipping points flagged	Amazon dieback + Polar ice + Coral die-off + North Atlantic circulation + Permafrost carbon
Living Planet Index coverage	~35,000 population trends for 5,495 vertebrate species

REVISE: *Shankar IAS / PMF IAS — Biodiversity threats chapter. NCERT Class XII Biology for ecosystem services.*
PYQ LINK: *UPSC 2016 asked about WWF. UPSC 2022 asked about Living Planet Index. Recurring report.*

SHOULD-KNOW

8.4 Environmental Performance Index (EPI) 2024 — India's ranking

Prelims · Environment + International Affairs · Also: State PCS

The Environmental Performance Index (EPI) — published by Yale University's Center for Environmental Law & Policy with Columbia's Earth Institute — ranks countries on 58 environmental performance indicators across 11 issue categories (climate change, environmental health, ecosystem vitality). EPI 2024 ranked India 176 out of 180 countries (score 27.6). India disputed this ranking, citing methodological concerns about indicator weights (particularly on projected emissions).

The Climate Change Performance Index (CCPI) — a separate index by Germanwatch, NewClimate Institute, and Climate Action Network — ranked India 10th in CCPI 2024 (released COP28) and 10th again in CCPI 2025. India's strong CCPI performance is due to low per-capita emissions and relatively strong renewable energy targets. The Environmental Performance vs Climate Performance gap reflects the tension between absolute emission levels and per-capita fairness.

KEY FACTS FOR PRELIMS

EPI	Yale University + Columbia University; released biennially
EPI 2024	India ranked 176/180 (score 27.6); disputed by India
EPI categories	3 policy objectives: Climate Change + Environmental Health + Ecosystem Vitality
CCPI	Germanwatch + NewClimate Institute + Climate Action Network; released annually at COP
CCPI 2024	India ranked 10th (released at COP28)
CCPI 2025	India ranked 10th again (released at COP29); strong on per-capita & renewables

Top 3 in CCPI 2025	Denmark, Netherlands, UK (top 3 ranked)
CCPI categories	GHG Emissions (40%) + Renewable Energy (20%) + Energy Use (20%) + Climate Policy (20%)

REVISE: *Shankar IAS / PMF IAS — Indices & Reports summary table. Keep EPI vs CCPI distinction handy.*
PYQ LINK: *UPSC 2022 asked about EPI. Rankings and methodologies are testable.*

SHOULD-KNOW

8.5 Protected Planet Report 2024 and Forest Declaration Assessment 2025

Prelims · Environment · Also: State PCS

The Protected Planet Report 2024 — published by UNEP-WCMC and IUCN, released at CBD COP16 Cali — tracks global progress on the 30×30 target (KMGBF Target 3). Key finding: 17.6% of land and inland waters and 8.4% of coastal and marine areas are under protection — well short of the 30% target by 2030. The report highlights the role of Other Effective Area-Based Conservation Measures (OECMs) — conservation outside formal protected areas — which now cover ~1% of land globally, up sharply.

The Forest Declaration Assessment 2025 (released before COP30) concluded that the world is off-track for the Glasgow Leaders' Declaration on Forests and Land Use (2021 COP26), which pledged to halt and reverse deforestation by 2030. Global deforestation in 2024 was 6.37 million hectares — 45% above the pathway needed. Tropical primary forests lost at unprecedented rates, with fires driving major losses. The report calls for strengthened NDCs and stronger implementation of forest-related commitments.

KEY FACTS FOR PRELIMS

Protected Planet Report 2024	Published by UNEP-WCMC + IUCN; released at CBD COP16 Cali
Global protected area	17.6% land & inland waters; 8.4% coastal & marine
30×30 target	KMGBF Target 3; 30% land and marine areas by 2030
OECMs	Other Effective Area-Based Conservation Measures — complement formal PAs
Forest Declaration Assessment 2025	Pre-COP30 release; world off-track for 2030 deforestation halt
2024 global deforestation	6.37 million hectares — 45% above pathway needed
Glasgow Leaders' Declaration on Forests	COP26 2021; 145+ countries pledged to halt/reverse deforestation by 2030
Main cause of tropical forest loss 2024	Fires — accounting for ~49% of primary forest loss

REVISE: *Shankar IAS / PMF IAS — Forests + Biodiversity chapters. Know each major report's publisher and its key metric.*
PYQ LINK: *UPSC 2019 asked about protected area targets. These reports are directly Prelims-relevant.*

GOOD-TO-KNOW

8.6 India's domestic reports — CSE State of Environment and State of Global Climate Report

Prelims · Environment · Also: State PCS

The State of India's Environment Report — published annually by the Centre for Science and Environment (CSE) and Down To Earth magazine — provides the domestic counterpart to international reports. SoE 2025 found that India had the 3rd-highest greenhouse gas emissions globally (after China and the US), but with per-capita emissions below the global average. The report also found that India's air pollution-related premature deaths crossed 1.67 million/year (WHO estimates).

The WMO Regional State of the Climate in Asia 2024 report (released April 2025) identified Asia as the world's most disaster-hit region by floods, storms, and landslides in 2024. India experienced record heatwaves in April–June 2024 (peak above 50°C in several places). India Meteorological Department (IMD) confirmed 2024 as India's warmest year on record since 1901, with annual mean temperature 0.65°C above long-term average.

KEY FACTS FOR PRELIMS

SoE Report	Centre for Science and Environment (CSE) + Down To Earth; annual
SoE 2025	India 3rd in absolute GHG emissions; below global avg per capita
Air pollution deaths	~1.67 million premature deaths/year (WHO)
State of Climate in Asia 2024	WMO; released April 2025; Asia most disaster-hit region
India 2024 heatwaves	April–June; peak above 50°C in several locations
India 2024 temperature	Warmest year since 1901 records; +0.65°C above long-term average (IMD)
CEEW	Council on Energy, Environment & Water (Delhi); key Indian think tank on climate
TERI	The Energy & Resources Institute (Delhi); founded 1974 by Darbari Seth; JRD Tata was first Chairman; R.K. Pachauri was first Director (1981–2015)

REVISE: *Shankar IAS / PMF IAS — Environment think tanks and institutions. Know CSE, CEEW, TERI distinctions.*
PYQ LINK: *UPSC 2024 asked about TERI and similar institutions. Source attribution questions are testable.*

SECTION 8 — PRACTICE

Test yourself

Q1. Match the following reports with their publishing organizations:

(a) Emissions Gap Report — UNEP
(b) Living Planet Report — WWF
(c) Environmental Performance Index — Yale University
(d) State of the Global Climate Report — Food and Agriculture Organization

How many of the above pairs are correctly matched?

(a) Only one
(b) Only two
(c) Only three
(d) All four

Q2. Consider the following statements regarding the Intergovernmental Panel on Climate Change (IPCC):

(a) It was established in 1988 jointly by the United Nations Environment Programme (UNEP) and the World Meteorological Organization (WMO).

(b) It conducts original climate research and publishes its findings as Assessment Reports.
(c) Its Sixth Assessment Report (AR6) was completed in 2023.
(d) It was awarded the Nobel Peace Prize in 2007.

Which of the statements given above are correct?

(a) 1, 3 and 4 only
(b) 1, 2 and 4 only
(c) 2 and 3 only
(d) All four

Q3. With reference to the Living Planet Report 2024 published by WWF, consider:

(a) The report found a 73% average decline in monitored wildlife populations between 1970 and 2020.
(b) Freshwater populations showed the steepest decline of around 85%.
(c) The Living Planet Index database is maintained by the Zoological Society of London.
(d) Latin America and the Caribbean showed the worst regional decline at around 95%.

Which of the statements given above are correct?

(a) 1 and 2 only
(b) 1, 2 and 3 only
(c) All four
(d) 3 and 4 only

Q4. Statement I: The Climate Change Performance Index (CCPI) consistently ranks India among the top 10 countries. Statement II: India's low per-capita emissions and relatively strong renewable energy targets contribute to its strong CCPI ranking.

(a) Both Statement I and Statement II are correct and Statement II is the correct explanation for Statement I
(b) Both Statement I and Statement II are correct but Statement II is not the correct explanation for Statement I
(c) Statement I is correct but Statement II is incorrect
(d) Statement I is incorrect but Statement II is correct

Q5. With reference to the Protected Planet Report 2024, consider:

(a) It is published by UNEP-WCMC in collaboration with IUCN.
(b) It was released at the CBD COP16 in Cali, Colombia.
(c) Approximately 17.6% of the world's land and inland waters are currently under protection.
(d) Other Effective Area-Based Conservation Measures (OECMs) are counted towards KMGBF Target 3.

Which of the statements given above are correct?

(a) 1 and 3 only
(b) 2 and 4 only
(c) 1, 2 and 3 only
(d) All four

ANSWERS WITH EXPLANATIONS

Q1. Answer: (c) Only three

First three pairs are correct. The State of the Global Climate Report is published by the WMO (World Meteorological Organization), not FAO. FAO publishes reports like the State of the World's Forests and State of the World's Biodiversity for Food and Agriculture.

Q2. Answer: (a) 1, 3 and 4 only

Statement 2 is incorrect — IPCC does NOT conduct original research. It reviews and synthesizes existing scientific literature. IPCC was set up 1988 by UNEP+WMO; AR6 completed 2023; Nobel Peace Prize 2007 (shared with Al Gore).

Q3. Answer: (c) All four

All four statements are correct. WWF's Living Planet Report 2024 confirmed 73% decline in monitored vertebrate populations (1970–2020); steepest in freshwater (85%); maintained by ZSL; worst in Latin America & Caribbean (95%).

Q4. Answer: (a)

Both statements are correct AND II correctly explains I. India ranked 10th in CCPI 2024 and 10th in CCPI 2025. The CCPI gives 40% weight to emissions, 20% to renewables, 20% to energy use, and 20% to policy — India's per-capita emissions (~2 t vs global avg ~6 t) and strong RE targets pull it high.

Q5. Answer: (d) All four

All four statements are correct. Protected Planet Report 2024 published by UNEP-WCMC + IUCN; released at CBD COP16 Cali; 17.6% land and inland waters protected; OECMs count toward 30×30 target under KMGBF Target 3.

SECTION T30

Top 30 Must-Know Facts

The exam-day crib sheet

The night before the exam, read only this. Thirty facts — drawn from the most likely-to-be-tested items across all eight sections. If you know these thirty, you have covered at least six of the projected environment questions on the paper.

01	**COP30 venue & outcome**	Belém, Brazil; Nov 2025; Global Mutirão Decision; USD 1.3 trillion/year by 2035; tripling adaptation finance; no fossil-fuel roadmap.
02	**COP31 host**	Antalya, Türkiye (Nov 2026). Australia withdrew bid in exchange for chairing talks.
03	**COP29 NCQG figure**	USD 300 billion/year by 2035 (developed→developing); India rejected as inadequate.
04	**CBD COP16 flagship**	Cali Fund for Digital Sequence Information (DSI); 0.1% revenue or 1% profits from large companies; 50% to IPLCs.
05	**UNCCD COP16 venue & pledge**	Riyadh, Saudi Arabia, Dec 2024; Riyadh Global Drought Resilience Partnership mobilized USD 12.15 billion.
06	**Ramsar sites (India)**	98 as of Jan 2026 — highest in Asia, 3rd globally. Tamil Nadu leads with 20.
07	**India's first Ramsar sites (1981)**	Chilika Lake (Odisha) + Keoladeo NP (Rajasthan).
08	**Montreux Record (India)**	Keoladeo NP + Loktak Lake (Manipur). Chilika removed in 2002.
09	**Tiger Reserves (total)**	58. 58th = Madhav, MP (March 2025); 57th = Ratapani, MP (Dec 2024); 56th = Guru Ghasidas-Tamor Pingla, CG (2024).
10	**Tiger census 2022**	3,682 wild tigers (75% of world). MP leads with 785. Jim Corbett has 231 (most in any reserve).
11	**Largest tiger reserve**	Nagarjunsagar-Srisailam (AP+Telangana, ~3,296 sq km).
12	**Smallest tiger reserve**	Bor (Maharashtra, 138 sq km).
13	**India's NDC (updated 2022)**	45% emissions-intensity cut below 2005 by 2030; ~50% non-fossil capacity by 2030; additional 2.5–3 Bt CO2e sink.
14	**Net-zero target**	2070 (announced COP26 Glasgow 2021 as part of Panchamrit).
15	**ISFR 2023 headline**	Forest + tree cover: 8,27,357 sq km = 25.17% of geographical area; mangroves: 4,992 sq km (down 7.43).
16	**Top 3 states (F&T cover)**	Madhya Pradesh > Arunachal Pradesh > Maharashtra. Top 3 (% forest): Lakshadweep > Mizoram > Andaman & Nicobar.
17	**Project Cheetah**	World's first intercontinental large-carnivore translocation. 57 cheetahs (2026). Sources: Namibia, South Africa, Botswana. NTCA implementing body.

18	**Gandhi Sagar WLS**	Second cheetah site (MP, not Rajasthan); relocation March 2025.
19	**Great Indian Bustard**	CR, <150 left. Thar Desert. SC modified undergrounding order (March 2024).
20	**KMGBF key targets**	Target 3 = 30×30 (conserve 30% land & marine by 2030). Target 19 = mobilize USD 200 billion/year.
21	**CCTS status**	Indian compliance carbon market. Intensity-based. 7 sectors notified (Oct 2025 + Jan 2026). Baseline FY2023–24. Trading by mid-2026.
22	**CCTS institutions**	BEE = Administrator; CERC = Regulator; Grid Controller of India = Registry. NSCICM oversees.
23	**Green Credit 2025 rules**	5-year restoration; 40% canopy density; non-tradable; one-time exchange only.
24	**PM Surya Ghar**	Launched 13 Feb 2024. Target: 1 crore rooftop solar households. Up to 300 free units/month. Rs 75,021 crore outlay.
25	**500 GW target**	Non-fossil capacity by 2030. ~230+ GW achieved by early 2026. Solar ~90 GW, Wind ~47 GW.
26	**ISA HQ**	Gurugram, Haryana — only intergovernmental body outside UN system HQ'd in India. India + France co-founders (2015).
27	**Bharat Small Reactor**	Announced Budget 2025–26. ~200 MWe indigenous SMR. PPP mode. Rs 20,000 crore nuclear corpus.
28	**IUCN 9 categories**	EX → EW → CR → EN → VU → NT → LC → DD → NE.
29	**WWF Living Planet 2024**	73% average decline in wildlife populations (1970–2020). Freshwater worst (85%).
30	**WMO 2024**	2024 was the warmest year on record; +1.55°C above pre-industrial (first time on annual basis).

> *If you only have thirty minutes on exam morning, read these thirty facts twice. If you have two hours, read them once and then skim the full magazine.*

SECTION GPT

Grand Practice Test

25 MCQs · One full section of Prelims-pattern questions

Complete this test in 30 minutes. Mark one option only. Note your answers. Check them against the answer key at the end. Target: 18 out of 25. If you score less, revise the topic-specific practice at the end of each section before the exam.

SECTION A — QUESTIONS

Attempt all 25

Q1. The flagship outcome of COP30 held in Belém, Brazil in November 2025 was:

(a) The Paris Agreement Implementation Protocol
(b) The Global Mutirão Decision
(c) The Baku Climate Accord
(d) The Just Transition Convention

Q2. The Cali Fund, launched in 2024–25, is related to:

(a) Ocean acidification research
(b) Benefit-sharing from Digital Sequence Information on genetic resources
(c) Financing of marine protected areas under BBNJ
(d) Carbon capture and storage in developing countries

Q3. As of January 2026, the total number of Ramsar sites in India is:

(a) 85
(b) 91
(c) 96
(d) 98

Q4. Which state has the highest number of Ramsar sites in India?

(a) Gujarat
(b) Tamil Nadu
(c) Karnataka
(d) West Bengal

Q5. India's 58th Tiger Reserve, notified in March 2025, is:

(a) Ratapani Tiger Reserve
(b) Madhav Tiger Reserve
(c) Guru Ghasidas-Tamor Pingla Tiger Reserve
(d) Veerangana Durgavati Tiger Reserve

Q6. Project Cheetah's second site in Madhya Pradesh, where cheetahs were relocated in 2025, is:

(a) Kanha National Park
(b) Panna Tiger Reserve
(c) Gandhi Sagar Wildlife Sanctuary
(d) Madhav National Park

Q7. Under India's updated Nationally Determined Contribution (2022), the emissions intensity of GDP is to be reduced by what percentage below 2005 levels by 2030?

(a) 33-35%
(b) 40%
(c) 45%
(d) 50%

Q8. India's net-zero target year, announced at COP26 Glasgow, is:

(a) 2050
(b) 2060
(c) 2070
(d) 2047

Q9. The Indian Carbon Credit Trading Scheme (CCTS) is administered by:

(a) Central Pollution Control Board
(b) Bureau of Energy Efficiency under Ministry of Power
(c) MoEFCC directly
(d) NITI Aayog

Q10. Which of the following is the baseline year for compliance targets under the CCTS?

(a) FY 2019-20
(b) FY 2020-21
(c) FY 2023-24
(d) FY 2024-25

Q11. The Green Credit Rules, 2025 stipulate that green credits for tree plantation are awarded only after:

(a) The trees have survived for two years with 20% canopy density
(b) The trees have survived for five years with 40% canopy density
(c) Ten trees have been planted per hectare on degraded land
(d) Payment of verification fees to ICFRE

Q12. The Kunming-Montreal Global Biodiversity Framework's 30×30 target is officially:

(a) Target 1
(b) Target 3
(c) Target 18
(d) Target 23

Q13. Which of the following Indian states has the highest number of Tiger Reserves as of March 2026?

(a) Karnataka

(b) Madhya Pradesh
(c) Maharashtra
(d) Uttarakhand

Q14. India's first and only Dugong Conservation Reserve is located in:
(a) Gulf of Kutch
(b) Gulf of Mannar
(c) Palk Bay, Tamil Nadu
(d) Lakshadweep

Q15. The total forest and tree cover of India as per the India State of Forest Report 2023 is:
(a) 21.76% of geographical area
(b) 24.62% of geographical area
(c) 25.17% of geographical area
(d) 33% of geographical area

Q16. According to ISFR 2023, which state has the largest forest and tree cover in absolute area?
(a) Arunachal Pradesh
(b) Madhya Pradesh
(c) Chhattisgarh
(d) Maharashtra

Q17. The Commission for Air Quality Management (CAQM) was established by an Act of Parliament in:
(a) 2019
(b) 2020
(c) 2021
(d) 2022

Q18. Under GRAP, Stage IV is triggered when the Air Quality Index (AQI) of NCR crosses:
(a) 300
(b) 400
(c) 450
(d) 500

Q19. The PM Surya Ghar: Muft Bijli Yojana was launched in:
(a) October 2023
(b) December 2023
(c) February 2024
(d) June 2024

Q20. The International Solar Alliance (ISA) has its headquarters in:
(a) New Delhi, India
(b) Paris, France
(c) Gurugram, Haryana
(d) Dubai, UAE

Q21. Which body is responsible for India's three-stage nuclear power programme?

(a) Ministry of New and Renewable Energy
(b) Ministry of Power
(c) Department of Atomic Energy
(d) Central Electricity Authority

Q22. As per the WMO State of the Global Climate Report released in 2025, the year 2024 was:

(a) Warmer than 2023 by a small margin
(b) The warmest year on record with global mean +1.55°C above pre-industrial
(c) Second warmest year on record after 2016
(d) Cooler than 2023 due to La Niña

Q23. The WWF Living Planet Report 2024 found that monitored wildlife populations have declined by what percentage between 1970 and 2020?

(a) 45%
(b) 60%
(c) 69%
(d) 73%

Q24. Cheetahs relocated to India since the launch of Project Cheetah have been brought from all the following countries EXCEPT:

(a) Namibia
(b) South Africa
(c) Kenya
(d) Botswana

Q25. The three Rio Conventions that emerged from the 1992 Earth Summit are:

(a) UNFCCC, CBD, Ramsar Convention
(b) UNFCCC, CBD, UNCCD
(c) UNFCCC, CITES, UNCCD
(d) CBD, CITES, Ramsar Convention

SECTION B — ANSWER KEY WITH EXPLANATIONS

Answer Key

Q1. Answer: (b) The Global Mutirão Decision
COP30's flagship outcome was the Global Mutirão Decision — 'Mutirão' is a Tupi-Guarani word meaning 'collective effort.' 29 decisions were adopted by consensus of ~200 Parties. Key provisions: USD 1.3 trillion/year finance by 2035; tripling adaptation finance by 2035; Belém Mission to 1.5°C; 59 Belém Adaptation Indicators.

Q2. Answer: (b) Benefit-sharing from Digital Sequence Information on genetic resources
The Cali Fund was launched at CBD COP16 (Cali 2024 / Rome Feb 2025) as the global fund for benefit-sharing from commercial use of DSI. Large companies contribute 0.1% revenue or 1% profits; at least 50% goes to Indigenous Peoples and local communities.

Q3. Answer: (d) 98
India had 98 Ramsar sites as of January 2026 — the highest in Asia and third-highest globally. Latest additions were Patna Bird Sanctuary (UP, 97th) and Chhari-Dhand (Gujarat, 98th).

Q4. Answer: (b) Tamil Nadu
Tamil Nadu leads with 20 Ramsar sites — the most of any Indian state.

Q5. Answer: (b) Madhav Tiger Reserve
Madhav Tiger Reserve in Shivpuri, Madhya Pradesh, was declared India's 58th Tiger Reserve on 10 March 2025. It covers 1,751 sq km and is formerly a royal hunting ground of the Scindia dynasty.

Q6. Answer: (c) Gandhi Sagar Wildlife Sanctuary
Gandhi Sagar WLS in north-western Madhya Pradesh (near the Rajasthan border, not in Rajasthan) was identified as the second cheetah site. Cheetahs were relocated there in March 2025. Note: Madhav National Park is a tiger reserve, not a cheetah site.

Q7. Answer: (c) 45%
India's updated NDC (August 2022) raised the emissions-intensity reduction target from 33–35% to 45% below 2005 levels by 2030. The non-fossil capacity target was also raised from 40% to ~50%.

Q8. Answer: (c) 2070
India announced net-zero by 2070 at COP26 Glasgow (November 2021) as part of the Panchamrit — one of five commitments.

Q9. Answer: (b) Bureau of Energy Efficiency under Ministry of Power
BEE is the Administrator of CCTS under the Ministry of Power. MoEFCC co-chairs the National Steering Committee (NSCICM). CERC regulates trading; Grid Controller of India is the Registry.

Q10. Answer: (c) FY 2023-24
FY 2023–24 is the baseline year. Compliance years are 2025–26 and 2026–27. Entities outperforming their targets earn Carbon Credit Certificates (CCCs); shortfalls must be bought or penalty paid.

Q11. Answer: (b) The trees have survived for five years with 40% canopy density
The 2025 revision made the scheme more rigorous — credits awarded only after 5 years (up from 2 years) based on canopy density of 40% (not tree count of 1,100/ha as earlier).

Q12. Answer: (b) Target 3
The 30×30 target — conserving 30% of land and marine areas by 2030 — is Target 3 of the 23 targets in the Kunming-Montreal Global Biodiversity Framework. Target 2 is ecosystem restoration; Target 18 is subsidies; Target 19 is finance.

Q13. Answer: (b) Madhya Pradesh

Madhya Pradesh leads all states with 9 Tiger Reserves — Kanha, Bandhavgarh, Pench, Satpura, Panna, Sanjay-Dubri, Veerangana Durgavati, Ratapani (57th, 2024), and Madhav (58th, 2025). MP also holds the highest tiger population of any state (785 tigers per 2022 census).

Q14. Answer: (c) Palk Bay, Tamil Nadu

India's first Dugong Conservation Reserve was notified on 21 September 2022 in Palk Bay, Tamil Nadu — covering about 448 sq km. The dugong is the only herbivorous marine mammal in India and is listed in Schedule I of the Wildlife Protection Act.

Q15. Answer: (c) 25.17% of geographical area

ISFR 2023 reported total forest and tree cover at 8,27,357 sq km = 25.17% of geographical area — up from 24.62% in 2021. Forest cover alone is 21.76%; tree cover alone is 3.41%.

Q16. Answer: (b) Madhya Pradesh

Madhya Pradesh leads with 85,724 sq km of forest and tree cover. Top three in order: Madhya Pradesh > Arunachal Pradesh > Maharashtra. (When considering forest cover alone without tree cover, Chhattisgarh is third after MP and Arunachal.)

Q17. Answer: (c) 2021

The Commission for Air Quality Management in the National Capital Region and Adjoining Areas was constituted by an Act of Parliament in 2021 (it replaced the Supreme Court-mandated EPCA). It is a statutory body with overriding powers.

Q18. Answer: (c) 450

GRAP Stage IV is triggered when AQI crosses 450 — the 'Severe+' category. Measures include ban on non-essential trucks, halt on construction, closure of schools for primary classes, and work-from-home advisory.

Q19. Answer: (c) February 2024

PM Surya Ghar: Muft Bijli Yojana was announced in the Interim Budget on 1 February 2024 and formally launched on 13 February 2024 by the Prime Minister.

Q20. Answer: (c) Gurugram, Haryana

ISA Secretariat is in Gurugram, Haryana — at the campus of the National Institute of Solar Energy (Gwal Pahari). It is the only intergovernmental organization outside the UN system headquartered in India.

Q21. Answer: (c) Department of Atomic Energy

The Department of Atomic Energy (DAE), which reports directly to the Prime Minister's Office, oversees India's three-stage nuclear programme through bodies like BARC, NPCIL, and IGCAR (Kalpakkam).

Q22. Answer: (b) The warmest year on record with global mean +1.55°C above pre-industrial

WMO's State of the Global Climate 2024 (released March 2025) confirmed 2024 as the warmest year on record, with global mean temperature approximately 1.55°C above pre-industrial for the first time on an annual basis.

Q23. Answer: (d) 73%

WWF Living Planet Report 2024 (released Oct 2024) found a 73% average decline in monitored wildlife populations between 1970 and 2020. Freshwater populations saw the steepest decline (85%); Latin America & Caribbean saw the worst regional decline (95%).

Q24. Answer: (c) Kenya

Cheetahs for Project Cheetah have been sourced from Namibia (2022), South Africa (2023), and Botswana (February 2026). Kenya has NOT been a source country. India initially attempted to get Asiatic Cheetahs from Iran but that did not materialize, and African Cheetahs from other African countries were pursued.

Q25. Answer: (b) UNFCCC, CBD, UNCCD

The three Rio Conventions are: UNFCCC (climate, 1992); CBD (biodiversity, 1992); UNCCD (desertification, 1994 — negotiated post-Earth Summit). Ramsar (1971, Iran) and CITES (1973, Washington) are NOT Rio Conventions, though often confused.

HOW DID YOU DO?

Scoring guide

Score	Assessment	Action
22–25	Excellent	You are exam-ready on environment.
18–21	Good	Target the sections where you scored less; revise the end-of-section practice.
13–17	Needs revision	Re-read the magazine and repeat this test after 2–3 days.
Below 13	Urgent	Schedule 3–4 days of dedicated environment revision using Shankar IAS / PMF IAS + this magazine.

BEYOND PREPARATION

UPSC preparation is hard. You are not alone.

BEYOND PREPARATION — I

Mental Health Support

The weight of UPSC preparation can be overwhelming. Years of study, the fear of failure, the pressure from family and society — it all adds up. If you are struggling with anxiety, depression, burnout, sleeplessness, or feelings of hopelessness, please know that seeking help is a sign of strength, not weakness. Your mental health matters — more than any exam.

At Neti Academy, we recognise that an aspirant's wellbeing is inseparable from their ability to prepare well. We do not claim to be mental health professionals. But we do commit to listening, to directing you toward qualified help, and to ensuring that no student of ours feels unseen.

If you need someone to talk to

- Please reach out to a qualified counsellor or therapist. Many government hospitals and medical colleges offer free or subsidised mental health consultations.
- iCall (TISS) — free confidential telephone counselling: 9152987821 (Mon–Sat, 8 am to 10 pm).
- Vandrevala Foundation Helpline — 1860 2662 345 (24/7, free).
- NIMHANS helpline — 080 46110007 (toll-free, 24/7).
- Tele-MANAS (Government of India) — 14416 (toll-free, 24/7 in multiple languages).

Everyday wellbeing for aspirants

- Keep a consistent sleep schedule. Seven hours is the floor, not the ceiling.
- Move your body every day. Thirty minutes of walking or yoga reduces anxiety more than any pep talk.
- Limit social media and comparison. Other aspirants' highlight reels are not your reality.
- Have at least one non-study relationship you invest in — a friend, a sibling, a parent.
- If you are having thoughts of self-harm, please call iCall or Vandrevala right now. Do not wait.

> *You are more than this exam. Your worth is not tied to your rank. Whatever happens on 24 May 2026, we are with you — before, during, and after.*

BEYOND PREPARATION — II

Plan B Support

UPSC is one of the most competitive examinations in the world — less than 0.2% of applicants reach the final list. Statistically, most serious aspirants will not clear CSE. This is not failure. This is mathematics.

A strong Plan B is not an admission of defeat. It is an insurance policy that lets you prepare for UPSC without the crushing fear of 'what if I don't make it.' Aspirants with a well-designed Plan B often perform better in UPSC itself, because the fear of catastrophic failure is removed.

What a realistic Plan B looks like

- **State PCS examinations (e.g., UPPCS, MPPCS, BPSC):** Syllabus overlap with UPSC is 70–85%. Your CSE preparation directly benefits these.
- **EPFO, SSC CGL, Bank PO, LIC AAO, IBPS:** Substantial syllabus overlap especially in GS paper areas; reasonable exam difficulty.
- **RBI Grade B, SEBI Grade A, NABARD:** Attractive alternatives for aspirants who enjoy economy and finance; competitive but manageable with structured prep.
- **Teaching and academia:** UGC NET is an immediate option; a Master's or PhD keeps options open.
- **Private sector domains that value policy knowledge:** Think tanks, policy research organisations, development sector, ESG/sustainability roles, journalism, and management consulting.

How to build a working Plan B

- Identify 2–3 non-UPSC exams or opportunities that align with your skills and degree.
- Spend 1–2 months working on a CV, resume, or alternative-path plan.
- Apply to at least one Plan B opportunity each year — treat it as non-negotiable.
- If you have cleared Prelims but not Mains, use the interview period to build Plan B connections.
- Your parents deserve to know you have a Plan B. It reduces their anxiety and yours.

> *A Plan B does not dilute your Plan A. It strengthens it. An aspirant who prepares without fear, with a financial and emotional safety net, prepares better.*

BEYOND PREPARATION — III

Mentorship

A good mentor compresses years of learning into months. They help you avoid the mistakes that every aspirant makes, they tell you the truth when everyone around you is offering platitudes, and they give you the structure your preparation needs. If you are preparing without mentorship, you are preparing alone — and UPSC is not an exam to face alone.

Neti Academy offers structured one-on-one mentorship. Our approach is different from most institutes. We do not sell you a course and disappear. We work with you, week by week, until you either clear the exam or decide to stop.

What our mentorship looks like

- Personalised study plan based on your current stage, strengths, and weaknesses.
- Weekly review calls — 30 minutes, focused on what to do next, not what you did wrong.
- Answer-writing evaluation with specific, actionable feedback — not just marks.
- Interview preparation by former civil servants and academic experts.
- Access to our curated test series aligned with the revised UPSC pattern.
- Psychological support across the full preparation cycle, not just exam time.

Who our mentorship is for

- First-time aspirants who want to build strong foundations.
- Second or third attempt aspirants who need to diagnose what is not working.
- Working professionals preparing alongside jobs who need efficient, structured guidance.
- Aspirants who have cleared Prelims but consistently struggle in Mains.

> *Good mentorship is not about more content. It is about the right content at the right time. To know more about our mentorship programmes, write to us at admin@netiacademy.com or visit www.netiacademy.com.*

CONNECT WITH US

Let us stay in touch

If this magazine helped you, we would love to hear from you. If it did not help, we would like to hear that even more — so we can do better in the next edition.

Reach us

- **Website:** www.netiacademy.com
- **Telegram channel:** t.me/netiacademy01
- **General queries and mentorship:** admin@netiacademy.com
- **Feedback on this magazine:** feedback@netiacademy.com

What you will find on our channels

- Daily current affairs — selected, summarised, prioritised.
- Weekly current affairs compilations and prelims-style practice questions.
- Revision guides, one-pagers, and topic summaries.
- Interview strategy content closer to the interview cycle.
- Answer-writing prompts and model answers for Mains preparation.

This is edition 1.0. Edition 1.1 will incorporate April–May 2026 current affairs as a supplement, released closer to the exam. Your feedback will shape what that edition looks like.

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